Chapter 5
pin Diodes

Most power diodes are pin-diodes, i.e. they possess a middle region with a much lower doping concentration than the outer p and n layers enclosing it. Compared with unipolar devices (see Chap. 6), pin-diodes have the advantage that the on-resistance is strongly reduced by high-level injection in the base region, which is known as conductivity modulation. Hence pin-diodes can be used up to very high blocking voltages. The base region is not intrinsic, as suggested by the name. The intrinsic case – doping in the range of $<10^{10}$ cm^{-3} – would not only be difficult to attain by technology, it would be even disadvantageous for the turn-off behavior and other properties. Power diodes usually have a $p^+n^-n^+$-structure, hence the so-called "i"-layer is actually an n^--layer. Since it is several orders of magnitude lower than the doping of the outer layers, the name pin-diode has become the usual denotation in almost every case.

From the viewpoint of application, power diodes can be distinguished in two main types:

Rectifier diodes for grid frequency of 50 Hz or 60 Hz: the switching losses play a subordinate role, and there is a high carrier lifetime in the middle layer.

Fast recovery diodes that work as freewheeling diodes for a switching device or that are in the output rectifier after a high-frequency transformer. They have to be generally capable of switching frequencies of up to 20 kHz and in switch-mode power supplies of 50–100 kHz and more. In fast diodes manufactured from silicon, the charge carrier lifetime in the middle low-doped layer has to be reduced to a defined low value.

5.1 Structure of the pin Diode

With respect to technology and doping profile, pin-diodes can be classified into two types. For pin diodes using epitaxial technology (epitaxial diodes, Fig. 5.1a), first, an n^--layer is deposited by epitaxy on a highly doped n^+-substrate. Then, the

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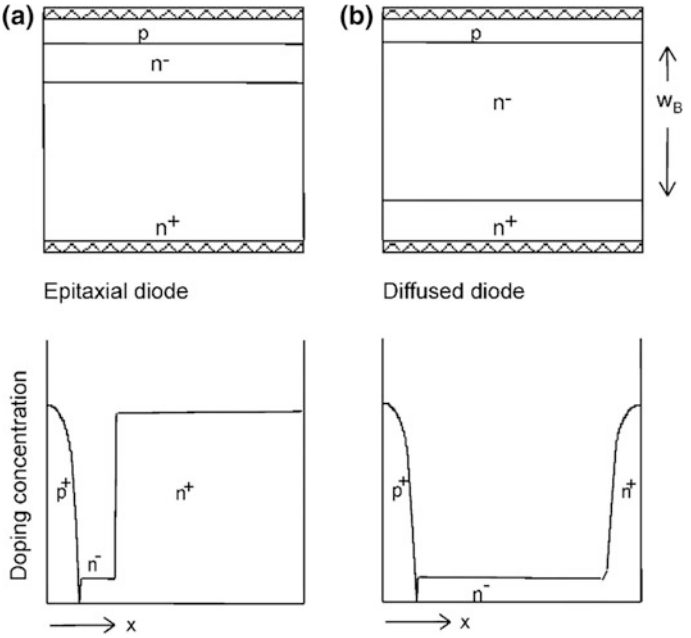


Fig. 5.1 Structure of pin power diodes. **a** Epitaxial diode. **b** Diffused diode

p-layer is diffused. With this process a very small base width w_B down to some micrometers can be created, whereby the silicon wafer is thick enough by the substrate to allow production with low wafer breaking and at high yield. By implementing recombination centers – in most cases by gold or platinum diffusion – very fast diodes can be realized. Since w_B is kept very small, the voltage drop across the middle layer is low. Epitaxial (epi-) diodes are mainly applied for blocking voltages between 100 and 600 V, however, some manufacturers also produce 1200 V with epi-diodes.

Because the costs of the epitaxy process are notable, diodes for higher blocking voltages – usually 1200 V and above – are fabricated by diffusion. For a diffused pin-diode (Fig. 5.1b), one starts with a low doped wafer in which the p^+-layer and the n^+-layer are created by diffusion. The thickness of the wafer now is determined by the thickness w_B of the middle n^--layer and the depths of the diffusion profiles. The required w_B is small for lower voltages. With deep n^+ and p^+-layers the wafer thickness can be increased again, but deep p-layers have disadvantages regarding the reverse-recovery behavior. The processing of such thin wafers is challenging. Infineon has introduced a technology for handling very thin wafers, down to a thickness of < 60 µm in the manufacturing process. With this technology, also freewheeling diodes for 600 V with shallow p and n^+ boarder layers can be fabricated as diffused diodes.

5.2 I–V Characteristic of the pin Diode

The I-V characteristic of a fast 300-V pin-diode measured at 25 °C as well as some definitions for parameters of the I–V characteristic are shown in Fig. 5.2. In the figure different scales in forward and reverse direction are applied. In forward bias, the characteristic associates a defined current I_F to the voltage drop V_F. This has to be distinguished from the maximally allowed voltage drop V_{Fmax} specified in manufacturers' data sheets. V_{Fmax} is the maximum forward voltage drop that can occur at a diode of this type under specified conditions. In most cases this value is significantly higher than the value measured for an individual sample due to tolerances of parameters during production, e.g. of the base width w_B. In fast diodes, the carrier lifetime strongly effects V_F. For fast gold- or platinum-diffused diodes of older generations, relatively high variations in the carrier lifetime are typical due to the difficulties to control these technologies, see Chap. 4. Some manufacturers also specify typical values, but this is not guaranteed for an individual sample.

In reverse bias, V_{BD} is the physical breakdown voltage of a given sample. I_R denotes the leakage current measured at a defined reverse voltage. This has to be distinguished from V_{RRM}, the maximal reverse voltage that is specified in the data sheet, as well as from I_{RM}, the specified maximal leakage current at V_{RRM}. Since the manufacturer takes into account data scattering and may add additional margins, at a single diode I_R cab be significantly lower and V_{BD} will be typically higher; the manufacture extends a warranty, however, only for V_{RRM} and I_{RM}, respectively.

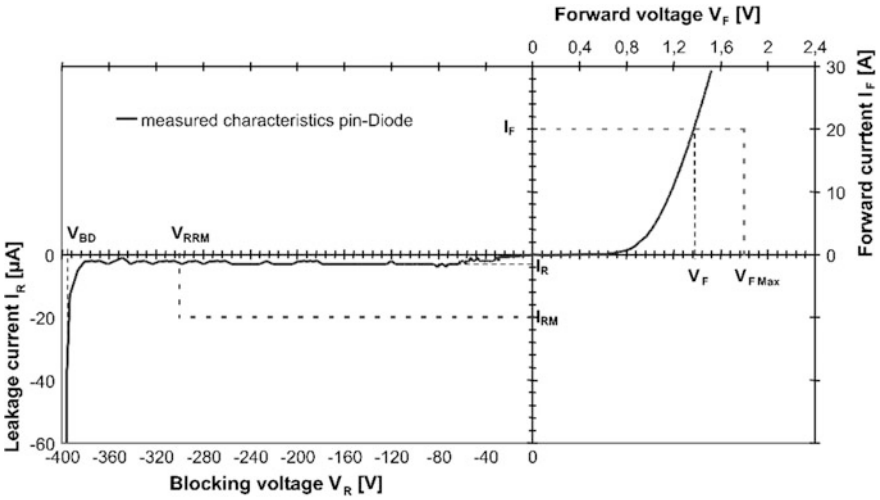


Fig. 5.2 I–V-characteristic of a fast pin-diode with some definitions of parameters

The I–V-characteristic of a diode is strongly temperature-dependent. With increasing temperature

- the leakage current I_R increases, I_R can be several orders of magnitude higher at the typical maximally allowed operation temperature of 150 °C than that at room temperature. Both, the diffusion component and generation component of the leakage current also increase; see Eq. (3.59) and Fig. 3.13
- the blocking voltage V_{BD} somewhat increases in accordance with the increase in breakdown voltage of avalanche breakdown; see Eq. (3.79) with the temperature dependent parameters in Eq. (3.90)
- the built-in voltage V_{bi} decreases, because according to Eq. (3.6), the determining parameter for the temperature dependency is the strongly temperature-dependent n_i^2. This corresponds to a decreased threshold voltage V_s. Also in the derivation of a threshold voltage from Eqs. (3.51) and (3.52), n_i^2 dominates the result.

5.3 Design and Blocking Voltage of the pin Diode

A dominating parameter for all characteristics of the diode is the width w_B of the low doped base-region. First of all, the base width together with the base doping concentration determines the blocking voltage. As illustrated in Fig. 5.3, different cases of the field shape during blocking near breakdown can be distinguished.

If w_B and N_D are chosen such that the space-charge region does not reach the n^+-layer (triangular field shape), the design is called a *Non-Punch-Through (NPT) dimensioning* [Bal87]. If w_B and N_D is chosen such that the space-charge penetrates the n^+-layer, then the field shape is trapezoidal or rectangular (Fig. 5.3b, c), and the structure is said to be of the *Punch-Through (PT)* type. The term 'punch-through' is used here in a different meaning as for thyristors. Whereas in the latter devices the space-charge region reaches up to a layer of opposite type of conductivity leading to breakdown, in diodes with PT-design the space charge layer

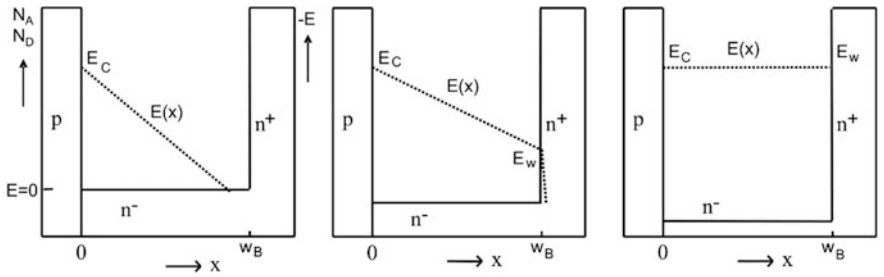


Fig. 5.3 Field distributions at breakdown in pin-($p^+n^-n^+$)-diodes with different design

is stopped by a highly doped region of the same type. The notation of diode design by NPT and PT is widely used. As shown by the area below the $|\mathsf{E}(x)|$-line, the PT-design yields a higher breakdown voltage for a given base width w_B, so it is enabling a smaller base width for a given V_{BD} than in the NPT-case. Fast pin power diodes of old generations have been dimensioned as NPT-diodes, because for a PT-design an acceptable reverse recovery behavior is more difficult to achieve. Nowadays, optimized fast diodes (see Sect. 5.7.4) use a PT-design to achieve low forward voltage drop and low stored charge. IGBTs are designed with trapezoidal field shape using a 'field stop layer' at the end of the n-base. This results in a significant reduction of the on-sate voltage drop. Also MOSFETs and Schottky diodes use mostly the PT-concept. In the present section we investigate how the base width necessary for a given blocking voltage in different cases of PT-design (Fig. 5.3b, c) compares with NPT-dimensioning. The p^+n- and nn^+-junctions are assumed to be abrupt. The power approach (3.75), $\alpha_{eff} = B|\mathsf{E}|^n$, is used with exponent $n = 7$, the constant B has then the value 2.11×10^{-35} cm^6/V^7 according to (3.83).

For NPT design, the relationships between breakdown voltage, base doping concentration and extension of the space charge region at breakdown have been given already in Sect. 3.3.2. Equation (3.85) says that width and doping density of the base region must satisfy for NPT the condition

$$w_B \geq w = \left(\frac{8}{B}\right)^{1/8} \cdot \left(\frac{\varepsilon}{q \cdot N_D}\right)^{7/8}$$

As condition, which w_B must satisfy for a given breakdown voltage V_{BD}, the inversion of (3.80) yields

$$w_B \geq w = 2^{2/3} \cdot B^{1/6} \cdot V_{BD}^{7/6} \tag{5.1}$$

pin-diodes with NPT-design in silicon must satisfy at 300 K these conditions. The NPT-case is assumed to include the equal sign, i.e. the case where the space-charge region just touches the n^+ layer without being stopped by it. If not stated otherwise, we are just dealing in what follows with the minimum base width $w_B = w$ satisfying the above conditions.

We consider now diodes with PT-dimensioning. Base width and doping density are then smaller than according to (5.1) and the preceding equation. The space-charge region penetrates the n^+-layer, where the field drops rapidly to zero as is depicted in Fig. 5.3b. To calculate the blocking voltage one has to determine first the critical field strength E_c, which can be different from that in the NPT case. The electric field over the base region at breakdown is given by

$$\mathsf{E}(x) = -\mathsf{E}_c + \frac{q \cdot N_D}{\varepsilon} \cdot x \tag{5.2}$$

Inserting this into $\alpha_{eff} = B \cdot |\mathsf{E}(\mathsf{x})|^7$ the breakdown condition (3.71) takes the form

$$\int_0^{w_B} B \cdot \left(\mathsf{E}_c - \frac{q \cdot N_D}{\varepsilon} \cdot x \right)^7 dx = 1 \tag{5.3}$$

The integration yields

$$\mathsf{E}_c^8 - \left(\mathsf{E}_c - \frac{q \cdot N_D}{\varepsilon} w_B \right)^8 = \frac{8 \cdot q \cdot N_D}{\varepsilon \cdot B} \tag{5.4}$$

Hence one can write:

$$\mathsf{E}_c = \left(\frac{8 q N_D}{B \varepsilon} + \mathsf{E}_w^8 \right)^{1/8} \tag{5.5}$$

with

$$\mathsf{E}_w = \mathsf{E}_c - \frac{q N_D}{\varepsilon} \cdot w_B \tag{5.6}$$

E_w represents the absolute value of the field strength at the nn$^+$ junction (see Fig. 5.3b). The implicit Eq. (5.5) for E_c can be solved by iteration, which except for very low $\mathrm{N_D}$ and $\mathsf{E}_w \approx \mathsf{E}_c$ converges rapidly. With E_c determined, the breakdown voltage is given by

$$V_{BD} = \frac{\mathsf{E}_c + \mathsf{E}_w}{2} = \left(\mathsf{E}_c - \frac{q N_D}{2\varepsilon} w_B \right) \cdot w_B \tag{5.7}$$

The breakdown voltage calculated in this way is plotted in Fig. 5.4 for a given base width (85 µm) as function of N_D (solid line). With decreasing N_D, the breakdown voltage increases monotonously. This holds also for other base widths.

Often one uses a design of moderate or weak punch-through with $\mathsf{E}_w \leq \mathsf{E}_c/2$. Then the term for E_w^8 in (5.5) is negligible and one obtains for the critical field.

$$\mathsf{E}_c = \left(\frac{8 \cdot q \cdot N_D}{B \cdot \varepsilon} \right)^{1/8} \tag{5.8}$$

This equation is identical with (3.78) for n = 7. From (5.7) and (5.8) the breakdown voltage is obtained as an explicit function of doping density and base width:

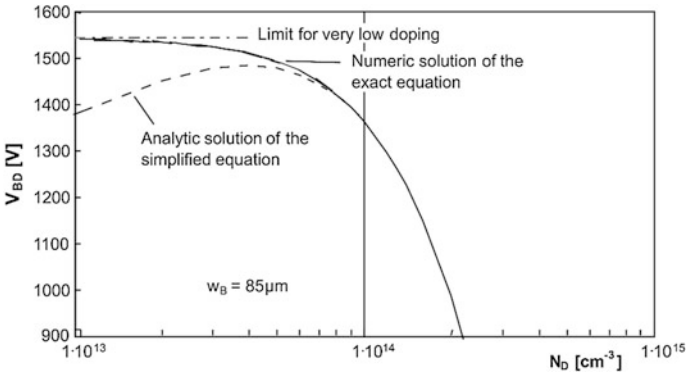


Fig. 5.4 Blocking voltage of a p^+nn^+ diode with given base width as a function of doping concentration of the base region

$$V_{BD} = \left(\frac{8qN_D}{\varepsilon B}\right)^{\frac{1}{8}} w_B - \frac{1}{2}\frac{qN_D}{\varepsilon} w_B^2 \tag{5.9}$$

This dependency on N_D for w_B = 85 µm is plotted in Fig. 5.4 too (dashed curve). At high N_D down to 8 × 10^{13} cm^{-3}, where $\mathsf{E}_w/\mathsf{E}_c = 0.51$, the curve coincides with the exact curve. Below this point the approximate solution shows a maximum, but becomes more and more incorrect. In some textbooks it is concluded from this approximation that the breakdown voltage has in fact a maximum at an optimal doping density and from this point *decreases* with decreasing doping. As is shown by the figure, this is an error. Neglecting the term E_w^8 in (5.5) is only allowed as long as doping is not too low.

The limit for very low doping can be derived immediately from the ionization integral. In this case, the field shape is rectangular, $\mathsf{E}_w = \mathsf{E}_c$, and the condition for avalanche breakdown (3.71) with the power approximation of ionization rates with n = 7 is

$$B \cdot \int_0^{w_B} \mathsf{E}_c^7 dx = 1 \tag{5.10}$$

Hence it follows $\mathsf{E}_c = 1/(Bw_B)^{1/7}$ and for $V_{BD} = \mathsf{E}_c \cdot w_B$

$$V_{BD} = \left(\frac{w_B^6}{B}\right)^{\frac{1}{7}} \tag{5.11}$$

This limiting value for the example w_B = 85 µm is also depicted in Fig. 5.4. The blocking voltage approaches this limit very quickly already for a doping in the range of 2 × 10^{13} cm^{-3}.

For the base width as function of V_{BD}, (5.11) leads to

$$w_B(PT, lim) = B^{\frac{1}{6}} \cdot V_{BD}^{\frac{7}{6}} \tag{5.12}$$

This lowest attainable base width is by a factor $2^{2/3}$ = 1.59 smaller than the minimum w_B for NPT-design given by (5.1):

$$w_B(PT, lim) = 2^{-\frac{2}{3}} w_B(NPT) \cong 0.63 \cdot w_B(NPT) \tag{5.13}$$

Although $w_B(PT,lim)$ is not half of $w_B(NPT)$, as would follow for equal critical field strength in both cases, the reduction by the PT-design is very significant.

Figure 5.5 shows the minimum width of the middle region for PT-design (5.12) and for NPT-design (5.1) as a function of breakdown voltage. The difference between w_B in the two cases results in a significant difference in forward voltage drop. For fast diodes with a rated voltage of 1200 V and with the necessary low carrier lifetime, the difference in V_F amounts up to 0.8 V. For higher voltages the reduction of forward voltage by the PT dimensioning is even larger. This is significant for the conduction losses. Consequently, a PT-dimensioning should be applied if possible. Nonetheless, this design also poses challenges, in particular regarding reverse-recovery behavior, as will be shown later.

A very high field at the nn$^+$-junction has on the other hand some disadvantages from a technological viewpoint. One drawback is that a field $\mathsf{E}_w \approx \mathsf{E}_c$ requires much more effort for edge termination at the surface. Therefore, only a moderate PT-dimensioning, often with $\mathsf{E}_w \leq {}^1/_2\, \mathsf{E}_c$ or smaller, is preferred. If the ratio $\mathsf{E}_w/\mathsf{E}_c$ is kept independent of the breakdown voltage, the dependency $w_B \sim V_{BD}^{7/6}$ is obtained for each value of $\mathsf{E}_w/\mathsf{E}_c$, where however the proportionality factor varies with the field ratio. For $\mathsf{E}_w/\mathsf{E}_c = 1/2$, the calculation yields

$$w_B(PT) = 0.70 \cdot w_B(NPT) \tag{5.14}$$

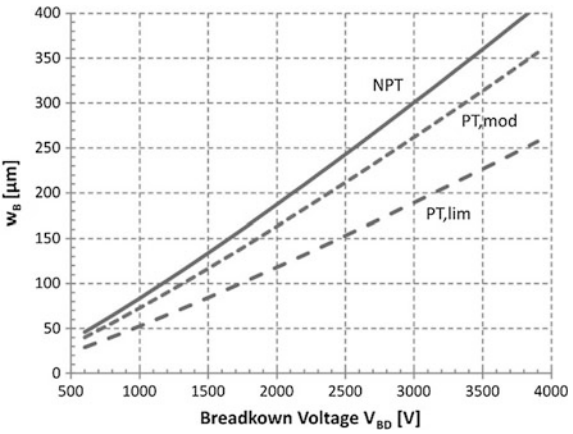


Fig. 5.5 Width w_B of the base region for dimensioning with triangular (NPT) and with rectangular field shape (PT,limit). The additional curve (PT,mod) refers to a moderate PT used for fast pin-diodes

In practice, tolerances in background doping and the adjusted base width implicate that the theoretical breakdown voltage is often not fully attained. Also the junction termination structure at the surface does not lead to 100% of the volume breakdown voltage in many cases. Hence the theoretical breakdown voltage must exceed the wanted maximum blocking voltage by a certain percentage. Taken from experience, an orientation value for a moderate PT-dimensioning can be given by:

$$w_B = \chi \cdot V_{BD}^{\frac{7}{6}} \quad \text{with} \quad \chi = 2.3 \times 10^{-6}\ \text{cmV}^{-\frac{7}{6}} \tag{5.15}$$

with a doping concentration chosen suitably somewhat below the NPT-value. The dependency (5.15) is depicted in Fig. 5.5 as the curve indicated with 'PT,mod'.

The described considerations on dimensioning are not only of relevance for diodes but also for other power devices whose base region contains a higher doped buffer layer to limit the extension of the space charge region. Particularly Schottky-diodes, MOSFETs and modern IGBTs use a PT-design. A discussion specifically on unipolar device design is given in Sect. 6.4.

A power device has to function over the temperature range from about 230 K to more than 400 K. The temperature dependence of V_{BD} is influenced by the PT-design as well, because it is determined besides the T-dependence of the critical field also by the manner how E_c enters the breakdown voltage. For NPT-design $V_{BD}(T)$ is described by (3.78) together with the temperature dependent parameters $n(T)$ and $B(T)$ as given by (3.87). For PT-design, we confine ourselves to the case $\mathsf{E}_w/\mathsf{E}_c \leq 1/2$. To get V_{BD} in dependence of temperature, Eq. (5.9) has to be used in a form which contains n and B as T–dependent variables. This is obtained by substituting E_c in (5.9) by the expression (3.78) which yields:

$$V_{BD} = \left(\frac{(n+1)qN_D}{\varepsilon B}\right)^{\frac{1}{n+1}} w_B - \frac{1}{2}\frac{qN_D}{\varepsilon} w_B^2 \tag{5.16}$$

$n(T)$ and $B(T)$ has to be inserted from (3.90). Since the critical field enters (5.16) only linearly, while for NPT-design V_{BD} is proportional to E_c^2 according to (3.79), Eq. (5.16) predicts a smaller variation with temperature than for NPT-layout. The cause is that the base width for PT-design is constant, whereas the width of the space-charge region in the NPT-case increases with T proportional to E_c.

Since the blocking capability must be given at the lowest operation temperature, the decrease of V_{BD} from room temperature downwards is of particular interest. Often the nn$^+$-junction is not abrupt, but the doping increases gradually. Then the measured temperature dependency is found to be between the predictions of (5.16) and (3.79).

Equation (5.16) can be used also for devices made of other semiconductors, if the PT-design with $E_w \lesssim {}^1/_2\, E_c$ is used and the parameters n and B are known. Thus the breakdown voltage of diodes in 4H–SiC-diodes can be calculated using the data for n und B given in Sect. 3.3.3.

5.4 Forward Conduction Behavior

5.4.1 Carrier Distribution

At forward bias, the lowly doped base zone of a pin-diode is flooded with carriers injected from the highly doped outer regions. The density of free carriers is increased by a few orders of magnitude above the background doping, and the conductivity in the low-doped zone is strongly enhanced or 'modulated'. Because in the neutrality condition $n = p + N_D^+$, the doping concentration N_D^+ is negligible, the hole and electron concentration in the base are approximately equal:

$$n(x) \approx p(x).$$

Figure 5.6 shows the carrier distribution $n = p$ in the middle region calculated for a 1200 V diode at a current density of 160 A/cm^2. Here, the emitter efficiency of highly doped regions has been assumed to be 1. n and p exceed the background doping by more than two orders of magnitude. The broken line is obtained if there would be equal mobilities for electrons and holes. Since the electrons in silicon are much more mobile than the holes, an asymmetric distribution, which is increased at the pn-junction, results.

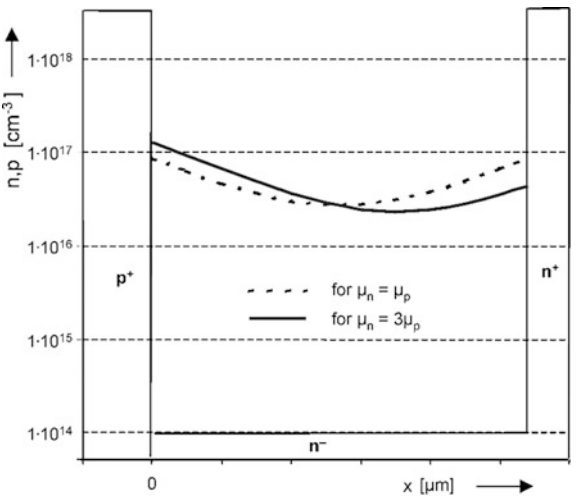


Fig. 5.6 Distribution of charge carriers in the base-region at forward conduction. Example of a diode with $\tau_{HL} = 0.48$ µs, $w_B = 108$ µm. The p$^+$- and n$^+$-layers are assumed to be ideal emitters

To calculate the carrier distribution, one starts with the transport Eqs. (2.43a) and (2.43b) which for $n = p$ take the form:

$$j_p = q \cdot \mu_p \cdot p \cdot \mathsf{E} - q \cdot D_p \cdot \frac{dp}{dx} \tag{5.17}$$

$$j_n = q \cdot \mu_n \cdot p \cdot \mathsf{E} + q \cdot D_n \cdot \frac{dp}{dx} \tag{5.18}$$

Since in the stationary case, which we consider, the total current density $j = j_n + j_p$ is independent of x according to (2.106), the electric field strength E is suitably expressed by j. Adding (5.17) and (5.18) one obtains

$$j = j_n + j_p = q \cdot (\mu_n + \mu_p) \cdot p \cdot \mathsf{E} + q \cdot (D_n - D_p) \cdot \frac{dp}{dx} \tag{5.19}$$

Hence, the electric field is

$$\mathsf{E} = \frac{\frac{j}{q} - (D_n - D_p) \cdot \frac{dp}{dx}}{(\mu_n + \mu_p) \cdot p} \tag{5.20}$$

The first term proportional to j is the resistive or ohmic field. The second field term determined by the concentration gradient and the difference in the diffusion constants is denoted as "Dember-field". The current caused by this field and the diffusion current in (5.19) add up to zero. Inserting Eq. (5.20) into (5.17) and (5.18), one obtains

$$j_p = \frac{\mu_p}{(\mu_n + \mu_p)} \cdot j - q \cdot D_A \cdot \frac{dp}{dx} \tag{5.21}$$

$$j_n = \frac{\mu_n}{(\mu_n + \mu_p)} \cdot j + q \cdot D_A \cdot \frac{dp}{dx} \tag{5.22}$$

where D_A is the following combination of the single diffusion constants:

$$D_A = \frac{2 \cdot D_n \cdot D_p}{D_n + D_p} \tag{5.23}$$

For this relationship, the Einstein relation $D_{n,p} = (kT/q) \cdot \mu_{n,p}$ has been used. D_A is called 'ambipolar diffusion constant', and the terms $\mp q D_A \cdot dp/dx$ are the ambipolar diffusion current densities. These include the current resulting from the Dember field, which makes the ambipolar diffusion current of electrons and holes oppositely equal. To obtain a differential equation for the carrier concentration, one

has to insert (5.20) or (5.21) into the relevant continuity equation. According to (2.103), the continuity equation for holes in the stationary case can be written

$$\frac{dj_p}{dx} = -q \cdot R = -q\frac{p}{\tau_{HL}} \tag{5.24}$$

where the recombination rate R is expressed according to Eqs. (2.49), (2.51) by p and the high level carrier lifetime τ_{HL}, neglecting the equilibrium carrier density. Inserting j_p from (5.21) the following differential equation for the carrier concentration is obtained:

$$D_A \cdot \frac{d^2p}{dx^2} = \frac{p}{\tau_{HL}} \tag{5.25}$$

where the mobility ratios and ambipolar diffusion constant are assumed to be constant. In what follows, also the lifetime τ_{HL} is assumed to be constant, which agrees with Eq. (2.74) of the SRH model. Then the differential Eq. (5.25) has the solutions $p(x) \propto \exp(\pm x/L_A)$, where L_A is the ambipolar diffusion length defined by

$$L_A = \sqrt{D_A \cdot \tau_{HL}} \tag{5.26}$$

In addition to the differential equation, the wanted carrier distribution has to satisfy conditions at the boundaries of the base region which are given by the emitter efficiency of the p^+ and n^+ region. In the present section, we assume that these regions are ideal emitters (with emitter efficiency 1), meaning that $j_n = 0$ at the p^+n junction and $j_p = 0$ at the nn^+-junction. Using these current densities in (5.21) and (5.22) one obtains the boundary conditions:

$$q\,D_p\frac{dp}{dx}(0) = -j/2, \qquad q\,D_n\frac{dp}{dx}(w_B) = j/2 \tag{5.26a}$$

To the same extent as $D_p < D_n$, the absolute value of the concentration gradient at $x = 0$ (p^+-side of the base region) is higher than at $x = w_B$ (n^+-side). Considering the simplified case $D_n = D_p = D_A = D$, the solution of (5.25) satisfying the conditions (5.26a) is

$$p(x') = \frac{jL_A}{2\,q\,D\,\sinh(w_B/(2 \cdot L_A))}\cosh\left(\frac{x'}{L_A}\right)$$

Here the coordinate x' has its origin in the middle of the base region: $x' = x - w_B/2$. The symmetrical carrier distribution in Fig. 5.6 (dashed line) was calculated using this equation together with the parameters given in the legend. The obtained cosh-distribution is well-known from the chainline. The smaller the ambipolar diffusion length L_A, the more pronounced is the sagging of the carrier distribution.

In the real case of different mobilities of electrons and holes, the carrier distribution can be written as

$$n(x') = p(x') = \frac{j \cdot \tau_{HL}}{2 \cdot q \cdot L_A} \left(\frac{\cosh \frac{x'}{L_A}}{\sinh \frac{w_B}{2 \cdot L_A}} - \frac{\mu_n - \mu_p}{\mu_n + \mu_p} \cdot \frac{\sinh \frac{x'}{L_A}}{\cosh \frac{w_B}{2 \cdot L_A}} \right) \tag{5.27}$$

It can be verified that this solution of (5.25) satisfies the boundary conditions (5.26a). The carrier distribution calculated from (5.27) for the given parameters is plotted in Fig. 5.6 as solid line. The concentration is at the pn-junction more than a factor 2 higher than that at the nn$^+$-junction (which later will prove to be a disadvantage for the reverse recovery behavior). The minimum has shifted towards the nn$^+$-junction. The term with $sinh(x'/L_A)$ in (5.27) expresses this asymmetry. Since for silicon $\mu_n \approx 3\mu_p$, the factor before this term consisting of the mobilities is approximately 0.5. As is noted, integration of (5.27) yields for the mean carrier concentration across the base:

$$\bar{n} = \bar{p} = \frac{1}{w_B} \int_{-w_B/2}^{w_b/2} p \cdot dx' = \frac{j \cdot \tau_{HL}}{q \cdot w_B}$$

For real p$^+$ and n$^+$-regions with emitter efficiencies < 1, the boundary conditions change and the factors in (5.27) are no longer valid. The carrier distribution in the base is written in this case suitably in the form:

$$p(x) = \frac{1}{\sinh(w_B/L_A)} \left(p_R \sinh(\frac{x}{L_A}) + p_L \sinh(\frac{w_B - x}{L_A}) \right) \tag{5.28}$$

p_L, p_R are the concentrations at the left and right edges of the base region and have to be determined from the general boundary conditions (see Sect. 5.4.5).

5.4.2 Junction Voltages

In a p$^+$nn$^+$-structure one has a space-charge region at the p$^+$n-junction and another at the nn$^+$-doping step, whereby both are connected with a built-in voltage V_{bi}(p$^+$n) and V_{bi}(nn$^+$), respectively. If a forward voltage V_F is applied, a part of it is used at the junctions to reduce the potential steps there and to raise the injected carrier densities in the base region similarly as for a single pn-junction. Additionally, the forward voltage provides for an ohmic voltage drop V_{drift} over the weakly doped base region needed for the current transport. Hence, if the junction parts are called V_j(p$^+$n), V_j(nn$^+$), one obtains:

$$V_F = V_j(p^+n) + V_{drift} + V_j(nn^+) \tag{5.29}$$

The internal voltage steps at the junctions

$$\Delta V(p^+n) = V_{bi}(p^+n) - V_j(p^+n),$$

$$\Delta V(nn^+) = V_{bi}(nn^+) - V_j(nn^+)$$

are related to the carrier concentrations at the neutral boundaries of the space-charge regions via Boltzmann factors. For the hole concentration p_L near the p^+n-junction in the neutral base and the electron density n_R at the n^+-side of the base (see Fig. 5.6), one obtains:

$$\frac{p_L}{p^+} = \exp\left(-\frac{q \cdot \Delta V(p^+n)}{kT}\right), \quad \frac{n_R}{n^+} = \exp\left(-\frac{q \cdot \Delta V(nn^+)}{kT}\right) \tag{5.30}$$

where the carrier densities p^+, n^+ of the highly doped regions are given by the doping densities N_A, N_D. Dividing the relationships (5.30) by the corresponding thermal equilibrium equations (see Chap. 3)

$$\frac{p_{n0}}{p^+} = \exp\left(-\frac{q \cdot V_{bi}(p^+n)}{kT}\right), \quad \frac{N_D}{n^+} = \exp\left(-\frac{q \cdot V_{bi}(nn^+)}{kT}\right),$$

one obtains for the external parts to the voltage drop at the junctions:

$$\begin{aligned} V_j(p^+n) &= \frac{kT}{q}\ln\frac{p_L}{p_{n0}} = \frac{kT}{q}\ln\frac{p_L \cdot N_D}{n_i^2} \\ V_j(nn^+) &= \frac{kT}{q}\ln\frac{n_R}{N_D} \end{aligned} \tag{5.31}$$

$V_j(p^+n)$ and $V_j(nn^+)$ differ significantly from one another depending on the doping density N_D of the base region. By leveling out the strong difference of built-in voltages, they make the internal voltage steps $\Delta V(p^+n)$ and $\Delta V(nn^+)$ more similar. The sum of both, the total external junction voltage V_j, does not depend on N_D:

$$V_j \equiv V_j(p^+n) + V_j(nn^+) = \frac{kT}{q}\ln\frac{p_L \cdot n_R}{n_i^2} \tag{5.32}$$

These equations hold independently of the injection level, but they will be used below for high-injection conditions where $p_L = n_L$ and $n_R = p_R$.

For the example of Fig. 5.6 with $p^+ = 2 \times 10^{18}$ cm^{-3}, $N_D = 7 \times 10^{13}$ cm^{-3} as doping of the middle layer, and $n^+ = 1 \times 10^{19}$ cm^{-3}, one obtains $V_{bi}(p^+n) = 0.721$ V,

$V_{bi}(\mathrm{nn}^+) = 0.307$ V. With p_L and p_R from the asymmetrical carrier distribution in Fig. 5.6, one attains $V_j(\mathrm{p}^+\mathrm{n}) = 0.654$ V, $V_j(\mathrm{nn}^+) = 0.161$ V, $V_j = 0.815$ V. For the total internal voltage steps, the results are $\Delta V(\mathrm{p}^+\mathrm{n}) = 0.067$ V, $\Delta V(\mathrm{nn}^+) = 0.136$ V.

5.4.3 Voltage Drop Across the Middle Region

Now the voltage V_{drift} that drops across the middle region has to be calculated. V_{drift} results by integration of the electric field which is given in Eq. (5.20). It contains in the nominator as second term the Dember field, which is proportional to the gradient of the carrier density. This term leads to the Dember voltage V_{Dem}. Using the Einstein relation (2.44), it follows from (5.20) that

$$V_{Dem} = \frac{kT}{q} \cdot \frac{\mu_n - \mu_p}{\mu_n + \mu_p} \cdot \ln \frac{p_L}{p_R} \tag{5.33}$$

For a symmetric carrier distribution this voltage term vanishes since $p_L = p_R$. But also for the actual asymmetrical distributions V_{Dem} is very small. For the example in Fig. 5.6, $V_{Dem} = 14.3$ mV results, which can be neglected.

From the term in Eq. (5.20) which is proportional to the current density, the voltage over the middle region is obtained as:

$$V_{drift} = \frac{j}{q \cdot (\mu_n + \mu_p)} \int_0^{w_B} \frac{1}{p(x)} dx \tag{5.34}$$

For a homogeneous distribution $p(x) = const$ the integral would be equal to w_B/p. For a not too strongly inhomogeneous distribution, this holds approximately if p is replaced by the average value $\bar{p}$. This leads to

$$V_{drift} = \frac{j \cdot w_B}{q \cdot (\mu_n + \mu_p) \cdot \overline{p}} \tag{5.35}$$

As can be verified by carrying out the integration in (5.34) with the exact carrier distribution (5.27) (see below), this is a good approach for $w_B/L_A \leq 3$, but can be used as a rough approximation up to $w_B/L_A = 4$. For $w_B/L_A = 3$ the error made with (5.35) is only 7%, although the inhomogenity is already considerable (cosh (1.5) = 2.35). The density of carriers represents a stored charge

$$Q_F = q \cdot A \cdot w_B \cdot \overline{p} \tag{5.36}$$

and, with this, it follows that

$$V_{drift} = \frac{I_F \cdot w_B^2}{(\mu_n + \mu_p) \cdot Q_F} \tag{5.37}$$

where $I_F = A \cdot j$ denotes the forward current (A area of the device).

5.4.4 Voltage Drop in the Hall Approximation

For the further calculation, we assume now as in Sect. 5.4.1, that recombination in the p^+ and n^+ region is negligible (emitter efficiency 1). This is suggested because the excess charge in the base is orders of magnitude higher than in the end regions (see Fig. 5.6). In the theory of diode characteristic this case is called Hall approximation [Hal52]. Since the minority carrier currents in the end regions are neglected, integration of the continuity Eq. (5.24) yields for the current density:

$$j = \frac{q \cdot w_B \cdot \bar{p}}{\tau_{HL}} \tag{5.38}$$

Multiplying (5.38) with $A \cdot \tau_{HL}$, one obtains for the stored charge:

$$Q_F = I_F \cdot \tau_{HL} \tag{5.38a}$$

Hence from (5.37) it follows that

$$V_{drift} = \frac{w_B^2}{(\mu_n + \mu_p) \cdot \tau_{HL}} \quad \text{for} \quad w_B/L_A < 3 \tag{5.39}$$

Expressing τ_{HL} by the ambipolar diffusion length L_A defined in (5.26) together with (5.23), Eq. (5.39) can be written in the form

$$V_{drift} = f(b) \cdot \frac{kT}{q} \left(\frac{w_B}{L_A}\right)^2 \quad \text{with} \quad f(b) = \frac{2\,b}{(1+b)^2} \tag{5.40}$$

where $b \equiv \mu_n/\mu_p$. Equation (5.40) can be found in many textbooks on device physics. The factor $f(b)$ is always $\leq ½$. For silicon, the mobility ratio is $b \approx 3$ which yields $f(b) \approx 3/8$. Like (5.37) Eqs. (5.39), (5.40) apply up to about $w_B/L_A = 4$. Even for $w_B = 4\ L_A$, V_{drift} amounts only to 0.16 V according to (5.40).

For a more pronounced sagging of the carrier distribution, the voltage drop is mainly produced by the region of lowest concentration, hence V_{drift} is underestimated by (5.39) and (5.40). The accurate formula for V_{drift} is obtained by carrying

out the integral in (5.34) with the exact distribution (5.27). With $d = w_B/2$ the result can be written as

$$V_{drift} = 4f(b) \cdot \frac{kT}{q} \cdot \sinh\left(\frac{d}{L_A}\right) \cdot \cosh\left(\frac{\Delta}{L_A}\right) \cdot \operatorname{arctg}\left(\frac{\sinh(d/L_A)}{\cosh(\Delta/L_A)}\right) \tag{5.41}$$

where Δ denotes the distance of the minimum of $p(x)$ from the middle of the base. The asymmetry enhances the voltage drop a little. In the limit $d/L_A \gg 1$, Eq. (5.41) turns into

$$V_{drift} = \pi f(b) \cdot \frac{kT}{q} \cdot \cosh\left(\frac{\Delta}{L_A}\right) \exp\left(\frac{d}{L_A}\right) \quad \text{for } w_B \gg 2L_A \tag{5.41a}$$

As mentioned, the equations of this section hold for the case of negligible recombination in the highly doped outer layers. However, the right hand side of (5.39) and (5.41) can be used essentially also in the general case as a factor in V_{drift}.

According to Eqs. (5.39) to (5.41), the voltage drop across the middle region does not depend (explicitly) on the current. The increase of current is neutralized by a proportional increase of the carrier concentration and so the ratio $j/\bar{p}$ in (5.35) is constant. Actually, the decrease of μ_n and μ_p causes an increase of V_{drift} with current. According to (5.38), the concentration $\bar{p}$ reaches very high values even at normal forward current densities. At such high concentrations, also the *lifetime* is significantly reduced because of Auger recombination. These effects together with the slight increase of the junction voltage according to Eq. (5.32) lead to an appreciable increase of the forward voltage with current. In the range where the Auger recombination predominates strongly and hence the lifetime in the base is very inhomogeneous, however, the analytical approach above becomes insufficient.

However, the application range of the equations is anyway restricted to much smaller concentrations and current densities. As mentioned, they are based on the condition of negligible recombination in the emitter regions, and this becomes significant at a decade smaller $\bar{p}$ than the Auger recombination. Typically the Hall approximation applies up to a current density of about 5 – 30 A/cm^2, depending on the lifetime and width of the base region. Above this limit the experimental forward voltage (see Fig. 5.2) is considerably higher and increases essentially stronger with current than calculated. This is supported by the observation that the stored charge increases essentially slower with current in the relevant range than according to the linear dependency (5.38a). In the next section, the theory will be expanded taking into account the emitter recombination, which resolves these discrepancies.

That the conductivity of the base region is enhanced very strongly by the high injection remains valid. However, due to this conductivity modulation, even high-voltage devices with their wide, weakly doped base region can be operated with high forward current densities without causing a very high forward voltage.

5.4.5 Emitter-Recombination, Effective Carrier Lifetime and Forward Characteristic

The influence of emitter recombination on the injection efficiency of pn-junctions has been described in detail in Sect. (3.4). Using (3.96) for the junctions of a forward biased pin-diode, a minority hole current $j_p(n^+) = q \cdot h_n \cdot p_R^2$ flows into the n^+-region, and an electron current $j_n(p^+) = q \cdot h_p \cdot p_L^2$ is flowing into the p^+-region, where h_p and h_n are the constants of the respective emitter region given by (3.107, 3.109). In the case of Fig. 5.7, these currents amount to $j_n(p^+) \approx 32$ A/cm^2 and $j_p(n^+) \approx 8$ A/cm^2, if h_p and h_n have a typical value of 2×10^{-14} cm^4/s (see Fig. 3.21). Hence, the current due to recombination in the end regions amounts to 25% of the base recombination current in this case where the lifetime in the base is very small (0.48 µs). For a higher τ_{HL}, higher current density or intentionally reduced emitter efficiency, the emitter recombination current can predominate.

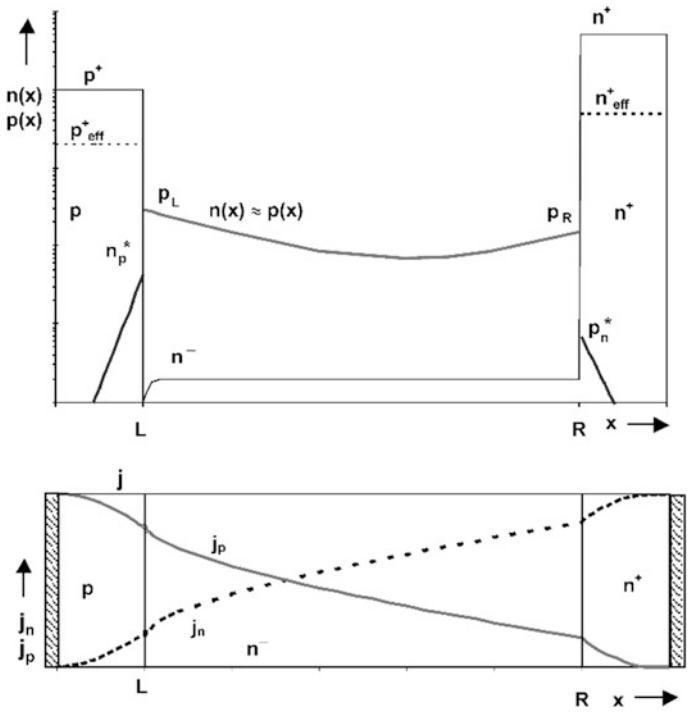


Fig. 5.7 pin-Diode with consideration of recombination in the border regions

To calculate the influence of emitter recombination on the forward characteristics of a pin-diode, we introduce an effective carrier lifetime τ_{eff} by [Sco69]:

$$\frac{\int_{-\infty}^{\infty} \Delta p dx}{\tau_{eff}} = \int_{-\infty}^{\infty} \frac{\Delta p}{\tau_p} dx \tag{5.42}$$

By this definition, τ_{eff} is a mean carrier lifetime of the structure including the emitter recombination. The integration extends from a point deep in the p^+-region ($x = -\infty$) over the base to a point deep in the n^+-layer ($x = \infty$). In the base where the injection level is high, the excess hole concentration Δp is equal to $p = n$, and in the p^+-region the recombination rate $\Delta p/\tau_p$ can be equated to the minority recombination rate $\Delta n/\tau_n(n^+) \approx n/\tau_p$. To realize the importance of the effective lifetime for device characteristics, we use the continuity Eq. (2.103) which in one-dimensional form can be written:

$$-\frac{\partial j_p}{\partial x} = q \cdot \frac{\Delta p}{\tau_p} + q \cdot \frac{\partial \Delta p}{\partial t} \tag{5.43}$$

Since the hole current deep in the p^+-region equals the total current ($j_p(-\infty) = j$), and deep in the n^+-region is zero ($j_p(\infty) = 0$), the integration of (5.43) yields

$$j = q \cdot \int_{-\infty}^{\infty} \frac{\Delta p}{\tau_p} dx + q \cdot \frac{d}{dt} \int_{-\infty}^{\infty} \Delta p \cdot dx, \tag{5.44}$$

Inserting (5.42) and multiplying with the area, one obtains

$$I = \frac{Q}{\tau_{eff}} + \frac{dQ}{dt} \tag{5.45}$$

where I denotes the current and Q the stored charge of excess carriers: $Q \equiv q \cdot A \cdot \int \Delta p \cdot dx$. (5.45) is a generally valid equation of charge dynamics. For a stationary forward current I_F, it takes the form:

$$Q_F = I_F \cdot \tau_{eff} \tag{5.46}$$

where Q_F is the stored charge for this special case. According to (5.46), the effective lifetime can be directly determined by measuring Q_F for a given forward current I_F. Compared with Eq. (5.38a), the lifetime τ_{HL} in the base is replaced in (5.46) by the effective lifetime of the structure. When the current is interrupted, τ_{eff} represents the decay time of the stored charge, since for $I_F = 0$ Eq. (5.45) yields $dQ/dt = -Q/\tau_{eff}$.

Immediately important for the I–V characteristic is that within the widely applicable approximation (5.35), (5.37), the effective lifetime determines the voltage drop across the middle region. By insertion of (5.46) in (5.37) one obtains:

$$V_{drift} = \frac{w_B^2}{(\mu_n + \mu_p) \cdot \tau_{eff}} \tag{5.47}$$

This equation is the generalization of (5.39) for the real case that the emitter efficiency of the junctions is below unity. Equation (5.47) is applicable if $w_B < 4 \cdot \sqrt{D_A \cdot \tau_{HL}}$ (see the discussion to Eq. (5.35)).

We evaluate the effective lifetime now in dependence on device parameters and on the stored charge or the mean concentration $\bar{p}$ in the base region. By splitting up the integration interval on the right hand side of (5.42) into the three neutral regions with constant lifetime one obtains (see Fig. 5.7).

$$\frac{1}{\tau_{eff}} \int_{-\infty}^{\infty} \Delta p \, dx = \frac{1}{\tau_n} \int_{-\infty}^{x_p} n \, dx + \frac{1}{\tau_{HL}} \int_{L}^{R} p \, dx + \frac{1}{\tau_p} \int_{x_n}^{\infty} p \, dx \tag{5.48}$$

The equilibrium minority carrier concentrations and likewise the contributions of the space charge layers from x_p to L and from R to x_n are neglected on the right hand side. Since the integrals are proportional to the respective stored charges, (5.48) can be written

$$\frac{Q}{\tau_{eff}} = \frac{Q_n(p^+)}{\tau_n(p^+)} + \frac{Q_B}{\tau_{HL}} + \frac{Q_p(n^+)}{\tau_p(n^+)} \tag{5.49}$$

where Q_B denotes the stored charge in the base, $Q_n(\mathrm{p}^+)$, $Q_p(\mathrm{n}^+)$ the stored charge of minority carriers in the p^+ and n^+ region, respectively, and $Q = Q_B + Q_n(\mathrm{p}^+) + Q_p(\mathrm{n}^+)$ the whole stored charge. Because of the low injection in the end regions and the relative small minority carrier diffusion length, the stored charges $Q_n(\mathrm{p}^+)$, $Q_p(\mathrm{n}^+)$ are small compared with the stored charge $Q_B = q \cdot w_B \cdot \bar{p}$, if the base width is not too small and the injection level not extremely high. Equation (5.49) shows that in spite of the small stored charges $Q_n(\mathrm{p}^+)$, $Q_p(\mathrm{n}^+)$, the *recombination* in the end regions can be significant if the lifetimes $\tau_n(\mathrm{p}^+)$, $\tau_p(\mathrm{n}^+)$ are correspondingly smaller than τ_{HL}. The latter can be caused by Auger recombination, a high density of recombination centers in the outer layers (see the discussion to Eq. (3.108)) or by a design of the end regions leading to high surface recombination.

Insertion of the exponential minority carrier distribution in the first and third integral on the right side of (5.48) yields the connection with the emitter parameters

introduced in Sect. 3.4. For the n^+-layer, we obtain with (3.43) and (3.104), (3.105) and neglecting the equilibrium density p_{n0}:

$$\frac{1}{\tau_p}\int_{x_n}^{\infty} p\,dx = \frac{L_p}{\tau_p}(n^+)\cdot p_n^* = \frac{L_p}{\tau_p}\cdot\frac{p_R^2}{n^+}e^{\Delta E_g/kT} = h_n\cdot p_R^2 \tag{5.50}$$

The bandgap narrowing ΔE_g results in an enhancement of the minority carrier concentration p_n^* and hence of the emitter parameter h_n. The analogous equation holds for the recombination integral over the p^+-region (first term on the right hand side of (5.48)):

$$\frac{1}{\tau_n}\int_{-\infty}^{L} n\,dx = \frac{L_n(p)}{\tau_n(p)}\cdot n_p^* = \frac{L_n}{\tau_n}\cdot\frac{p_L^2}{p^+}e^{\Delta E_g/kT} = h_p\cdot p_L^2 \tag{5.51}$$

If the integral on the left side in (5.48) is approximated by $w_B\cdot\bar{p}$, neglecting the stored minority carrier charges in the end regions, one obtains from (5.48), (5.50) and (5.51)

$$\frac{1}{\tau_{eff}} = \frac{1}{\tau_{HL}} + h_p\cdot\frac{p_L^2}{w_B\cdot\overline{p}} + h_n\cdot\frac{p_R^2}{w_B\cdot\overline{p}} \tag{5.52}$$

To correlate the mean concentration $\bar{p}$ with the concentrations p_L and p_R at the boundaries, the carrier distribution is used in the form (5.28). One obtains

$$\bar{p} = \frac{1}{w_B}\int_{L}^{R} p\,dx = \frac{L_A}{w_B}\cdot\tanh\left(\frac{d}{L_A}\right)\cdot(p_L+p_R) \tag{5.53}$$

For simpler writing we use again the letter d for $w_B/2$. Using (5.53), Eq. (5.52) can be written

$$\frac{1}{\tau_{eff}} = \frac{1}{\tau_{HL}} + \frac{H}{d}\cdot\left(\frac{d/L_A)}{\tanh(d/(L_A))}\right)^2\cdot\bar{p} \tag{5.54}$$

where

$$H = 2\cdot\frac{\eta^2 h_p + h_n}{(\eta+1)^2} \tag{5.55}$$

with $\eta = p_L/p_R$. The quantity H is to a sufficient approximation often independent of $\bar{p}$. For a symmetrical distribution ($\eta = 1$), H reduces to $(h_n + h_p)/2$. With a typical H-value of 2×10^{-14} cm^4/s, one obtains from (5.54) for a base width $w_B = 200$ µm at $\overline{n} = \overline{p} = 1\times10^{17}$ cm^{-3}:

$$\frac{1}{\tau_{eff}} = \frac{1}{\tau_{HL}} + \frac{1}{5.0\ \mu s}\left(\frac{d/L_A}{\tanh(d/L_A)}\right)^2 \quad \text{at} \quad \bar{p} = 1 \times 10^{17}\ \text{cm}^{-3}$$

Even if the carrier lifetime τ_{HL} is very high, the effective carrier lifetime remains below 5 µs in this case, the value given by the recombination in the boarder regions for $w_B/L_A \to 0$.

Measurements of the effective lifetime τ_{eff} and the high-level lifetime τ_{HL} in the base are plotted in Fig. 5.8 as functions of the mean concentration $\bar{n} = \bar{p}$ in the base. $\tau_{eff} = Q_F/I_F$ was determined by measuring the stored charge Q_F, which simultaneously delivers the concentration $\bar{n}$. The base lifetime τ_{HL} was determined from the carrier distribution measured via the recombination radiation profile. While τ_{HL} is found to be nearly independent of the carrier concentration, τ_{eff} decreases nearly by an order of magnitude in the shown range. This is in agreement with the above theory; particularly the variation of τ_{eff} with $\bar{n}$ can be described by (5.52) and (5.54). A point not in agreement with these equations is that τ_{eff} does not tend to the base lifetime τ_{HL} for small $\bar{n}$, but remains considerably smaller. We will come back to this effect at the end of this paragraph.

The boundary concentrations p_L, p_R depend strongly on the parameters h_p, h_n. In some modern fast power diodes, the injection efficiency of the p-emitter is made small while the n^+-region keeps the usual form with a small parameter h_n. This leads to an inversion of the carrier distribution compared with Fig. 5.6 and by means of this to an improved reverse recovery behavior. The influence of the

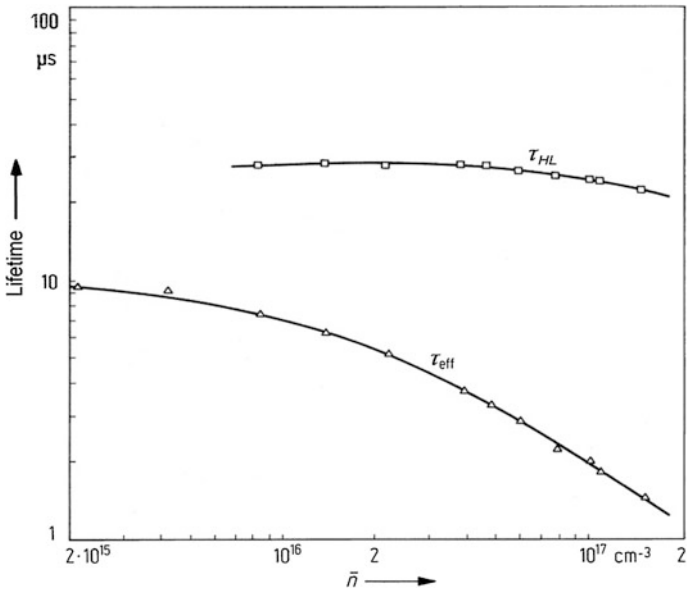


Fig. 5.8 Effective lifetime τ_{eff} of a forward biased pin-diode and lifetime τ_{HL} in the base region as functions of the mean carrier concentration in the base. See text. From [Sco82]

h-values on the carrier distribution follows from the continuity of the electron current at the p^+n-junction and the hole current at the nn^+-junction. Using (5.21), (5.22) together with (5.28) the boundary conditions can be written:

$$\frac{D_n}{D_n+D_p}j - q\frac{D_A}{L_A}\left(\frac{p_L}{\tanh(w_B/L_A)} - \frac{p_R}{\sinh(w_B/L_A)}\right) = q\,h_p p_L^2 \tag{5.56a}$$

$$\frac{D_p}{D_n+D_p}j + q\frac{D_A}{L_A}\left(\frac{p_L}{\sinh(w_B/L_A)} - \frac{p_R}{\tanh(w_B/L_A)}\right) = q\,h_n p_R^2 \tag{5.56b}$$

Without solving these equations explicitly for p_L and p_R, one can see, that p_L becomes smaller if h_p is enhanced. In the limit of very large emitter recombination (right hand side) compared with the recombination in the base, the ratio $\eta = p_L/p_R$ tends to $(D_n h_n/(D_p h_p))^{1/2}$, as follows by division of (5.56a) and (5.56b). A calculation of η as a function of $\bar{p}$ and the device parameters h_p, h_n, w_B and L_A is given in [Sco69]. The quantity H, which according to (5.54) determines the effective lifetime and hence the voltage drop V_{drift}, varies less than h_p because η decreases with increasing h_p. A measured carrier profile for a 1200-V diode using the mentioned design principle is shown later in Fig. 5.33 in Sect. 5.7.4. In this diode the p-emitter layer had a doping $N_A = 5 \times 10^{16}$ cm^{-3} and a thickness $w_p =$ 2 µm, which according to Eq. (3.106) results in $h_p = 2.6 \times 10^{-12}$ cm^4/s. The n^+-region had a doping concentration of about 10^{19} cm^{-3} and an estimated h_n-value of 2×10^{-14} cm^4/s. As shown by the figure, the boundary concentration p_L is about a factor 4 smaller than p_R. However, the recombination term $h_p p_L^2$ in (5.52) is still seven times higher than $h_n p_R^2$.

From (5.46) and (5.54), the current density can be expressed as a function of the concentration $\bar{p}$:

$$\begin{aligned} j &= \frac{Q_F/A}{\tau_{eff}} = \frac{q\cdot w_B\cdot\bar{p}}{\tau_{eff}} \\ &= q\cdot\bar{p}\cdot\left(\frac{w_B}{\tau_{HL}} + 2H\cdot\left(\frac{d/L_A}{\tanh(d/L_A)}\right)^2\cdot\bar{p}\right) \end{aligned} \tag{5.57}$$

The voltage drop across the base follows by insertion of (5.54) into (5.47)

$$V_{drift} = \frac{w_B}{\mu_n+\mu_p}\cdot\left(\frac{w_B}{\tau_{HL}} + 2H\cdot\left(\frac{d/L_A}{\tanh(d/L_A)}\right)^2\cdot\bar{p}\right) \tag{5.58}$$

The junction voltage (5.32) can be written using (5.53):

$$V_j = 2\,\frac{kT}{q}\ln\left(\frac{2\sqrt{\eta}}{1+\eta}\cdot\frac{d/L_A}{\tanh(d/L_A)}\cdot\frac{\bar{p}}{n_i}\right) \tag{5.59}$$

where as before $\eta = p_L/p_R$. With (5.57) to (5.59) the current-voltage characteristic is given in a parameter representation $j(\bar{p})$, $V_F(\bar{p}) = V_j(\bar{p}) + V_{drift}(\bar{p})$. This form of the characteristic has the advantage, that also the mobilities and ambipolar diffusion constant as well as the lifetime in L_A can be inserted as functions of $\bar{p}$ to consider carrier-carrier scattering and, if necessary, Auger recombination.

Forward characteristics calculated with these equations for different values of h_p, h_n are shown in Fig. 5.9 (solid curves). The dimensions of the base region are suitable for a 3 kV-diode. Carrier-carrier scattering is taken into account as described in Chap. 2. The neglect in the step from (5.34) to (5.35) is abandoned using the inhomogeneity factor given by the later Eq. (5.63). In the case $h_n = h_p = 2 \times 10^{-14}$ cm^4/s a realistic characteristic for diodes with these dimensions is obtained. For $h_n = h_p = 0$ (Hall case), the concentration $\bar{p}$ runs so high at high current densities that Auger recombination in the base becomes very significant. Within the above model, Auger recombination is considered replacing $1/\tau_{HL}$ by $1/\tau_{HL} + (c_{A,n} + c_{A,p}) \cdot \bar{p}^2$ [see Eq. (2.59)]. To test the accuracy of the model, also characteristics calculated numerically are plotted in Fig. 5.9 (dotted lines). They were obtained by determining the carrier distribution for each current density from the differential Eq. (5.25) using a lifetime including Auger recombination and variable mobilities in (5.21); the mobility sum in (5.34) was pulled under the integral. As is seen, the analytical approach is fairly accurate. Above the current range of the figure and in the Hall case, the analytical model is no longer suited. However, the Hall case is anyway hypothetical.

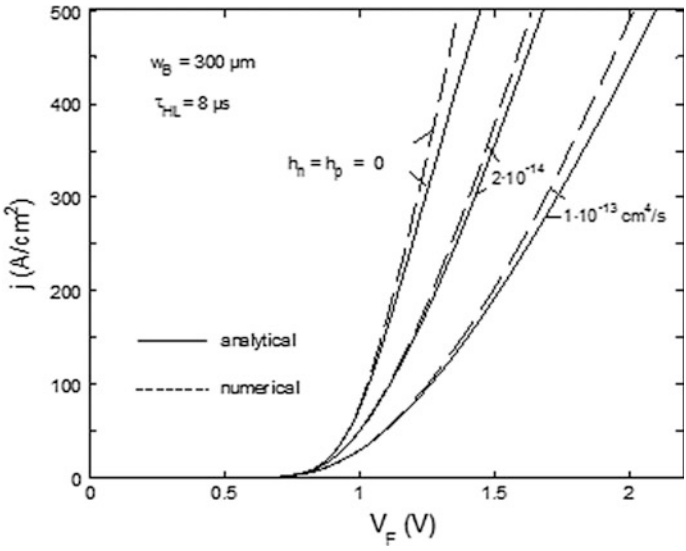


Fig. 5.9 Current voltage characteristics calculated with different emitter parameters

Sometimes the direct relationship between forward voltage and current density is desirable. To derive it, Eq. (5.57) is resolved first for $\bar{p}$, which yields

$$\frac{1}{\bar{p}} = \frac{q\, w_B}{2\, j\, \tau_{HL}} \left(1 + \sqrt{1 + \frac{2\, \tau_{HL}\, H}{q\, D_A\, \tanh^2(d/L_A)} \cdot j} \right) \tag{5.60}$$

Inserting this into (5.35) one obtains for the voltage drop across the middle region:

$$V_{drift} = \frac{w_B^2}{2\, (\mu_n + \mu_p)\, \tau_{HL}} \left(1 + \sqrt{1 + \frac{2\, \tau_{HL}\, H}{q\, D_A\, \tanh^2(d/L_A)} \cdot j} \right) \tag{5.61}$$

The junction voltage as a function of current density is obtained by insertion of (5.60) into (5.59). Hence the forward voltage $V_F = V_j + V_{drift}$ is given now as a function of the current density j.

Contrarily to the Hall approximation (5.39), the voltage drop V_{drift} depends now explicitly on the current density. For a sufficiently small emitter quantity H or small current density j, (5.61) reduces to (5.39). Since H cannot be made smaller than about 1×10^{-14} cm^4/s, however, the current dependent term in (5.61) becomes soon significant. The reference current density

$$j_0 \equiv \frac{q\, D_A\, \tanh^2(d/L_A)}{2\, \tau_{HL}\, H}$$

around which the current density j comes into play, is fairly small. If for example $H = 2 \times 10^{-14}$ cm^4/s, $d = 100$ µm, $\tau_{HL} = 4$ µs, $D_A = 15$ cm^2/s, j_0 amounts only to 11.1 A/cm^2. If H is made large for above mentioned reasons, j_0 will be still smaller. For $j \gg j_0$, (5.61) turns into

$$V_{drift} = \frac{w_B^2}{(\mu_n + \mu_p)\, L_A\, \tanh(d/L_A)\, \sqrt{2}} \cdot \sqrt{H \cdot j/q} \tag{5.62}$$

V_{drift} is here proportional to the square root of the current density, as far as the variation of the mobilities can be neglected. $\bar{p}$ is proportional to $\sqrt{j}$ in this case according to (5.60), hence $j/\bar{p}$ in (5.35) is also proportional to $\sqrt{j}$. This relationship is in accordance with measurements, which often yield a dependency $V_F - V_j \propto \sqrt{j}$ for power diodes.

Figure 5.10 shows the measurement of the forward characteristics of a fast 600-V diode and the result of a calculation with Eqs. (5.59)–(5.61). For the dependency of μ_n, μ_p, D_A on the carrier concentration, the Equations given in Sect. 2.6 and Appendix A are used. The calculation agrees with the measurement within a wide range of the current density.

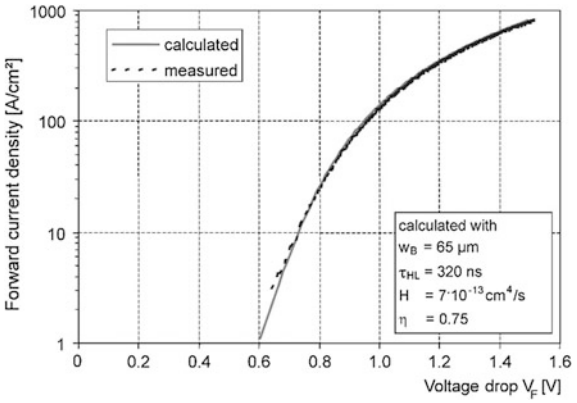


Fig. 5.10 Measured forward characteristics of a fast 600-V diode compared to calculation results with consideration of recombination in the emitter regions

If the carrier distribution is very inhomogeneous, i.e. very unsymmetrical or if it is characterized by a $d/L_A > 2$, the approximations (5.35) or (5.47), underlying our calculation, underestimates the voltage drop V_{drift} considerably. Carrying out the integral in (5.34) with the exact carrier distribution $p(x') = \cosh((x' - \Delta)/L_A)$ results in a voltage drop V_{drift} which has been given for the Hall case by (5.41). According to this equation, V_{drift} is by the factor

$$f = \left(\frac{L_A}{d}\right)^2 \cdot \sinh\left(\frac{d}{L_A}\right) \cdot \cosh\left(\frac{\Delta}{L_A}\right) \cdot arctg\left(\frac{\sinh(d/L_A)}{\cosh(\Delta/L_A)}\right) \tag{5.63}$$

higher than given by the approximation (5.40) (Δ denotes the distance of the minimum of $p(x)$ from the middle of the base). Since the carrier distribution has the same general *cosh*-form also if emitter recombination is present, the inhomogeneity factor (5.63) applies generally and has to be added correctly to the V_{drift} expressions (5.47), (5.58), (5.61) and (5.62). In Figs. 5.9 and 5.10 the factor has been taken into account.

The above equations can be used also for IGBTs to describe the I–V-characteristics in the saturated current range. The conduction behavior of several types of IGBTs is mainly determined by the p-emitter, since very shallow and low doped emitter regions are implemented. An IGBT-characteristic will be shown later in Fig. 10.2. In the range above the rated current one finds often a nearly linear or resistive characteristic.

In Fig. 5.8 it is observed that τ_{eff} for small $\bar{n}$, in the range of 10^{15} cm^{-3}, is nearly constant like the lifetime τ_{HL} in the base, but is essentially smaller than τ_{HL}. This result has been found for all samples till now. It has led to the conclusion that there is a linear recombination part entering the stored charge, which is not located in the volume of the base but must be located in the emitters or thin boundary layers of the base not resolved by the radiation profiles [Sco79, Coo83]. This recombination part called 'linear emitter recombination' has been mentioned already in Sect. 3.4 and is expressed by Eq. (3.112). In the later Fig. 5.33 in Sect. 5.7.4, the phenomenon appears again: The lifetime in the volume of the base is higher than given by Q_F/I_F

at small $\bar{p}$ where the quadratic emitter recombination is negligible. The sagging of the concentration towards the middle of the base is therefore smaller. For the incorporation of this effect into the theory of the I–V-characteristic we refer to the literature [Sco79].

Above calculations can be used to implement the parabolic shape of the forward characteristics of modern fast diodes in a circuit simulator. Such simulators need simplified models for the forward characteristics of diodes, and this description by a parabola is in most cases much closer to reality than the description with the Hall approximation, or the often used oversimplified description with a straight line.

5.4.6 Temperature Dependency of the Forward Characteristics

For low current, the forward voltage decreases with temperature, because the part of the voltage that drops at the pn-junction, i.e. the term $V_j(p^+n)$ given in Eq. (5.31), decreases with increasing n_i^2. The diode at low-current condition can thus be used as a temperature sensor (see Fig. 3.12 and Fig. 11.19 in Chap. 11). At increased current density, the temperature dependency of V_{drift} predominates and here, opposing effects have to be considered:

- The mobilities decrease with temperature, (see Fig. 2.14 and Appendix A1). According to (5.47), this leads to an increase in V_{drift}.
- The carrier lifetime increases with increasing temperature which, in turn, leads to a decrease in V_{drift}.

Since both effects are opposing, the resulting behavior depends on the specific technology, especially of the temperature dependency of the effect of the used recombination centers.

By using radiation-induced recombination centers, a curve as is depicted in Fig. 5.11 on the right-hand side is measured. The lines for the characteristics at 25 and 150 °C intersect at 150–200 A/cm^2, a current density which is typical for rated current of a 1200 V fast diode.

Even though an intersection point is likewise found for diodes without recombination centers (rectifier diodes for grid frequency) and for diodes using gold as recombination center, this intersection occurs for those devices typically at a 3-times higher current density.

By using platinum combined with a not very weakly doped p-emitter, no intersection point is found in the relevant current range. An example is given in Fig. 5.11 (left). In this case, the forward voltage strongly decreases with temperature. This is advantageous in terms of conduction losses, but this temperature behavior is very unfavorable for parallel connection of diodes. As the result of

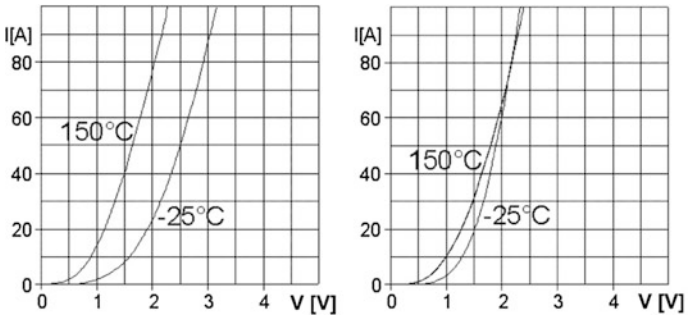


Fig. 5.11 Forward characteristics of fast 1200 V diodes and its temperature dependency. Left: Platinum-diffused diode. Right: Diode with radiation-induced recombination centers (CAL diode). Active area 0.32 cm^2

tolerances in the manufacturing process, there is always some variation of the forward voltage for different samples. In case of parallel connection, the diode with the lower voltage drop will attract more current. Consequently, it will generate higher conduction losses, its temperature will increase. That will result in a further reduction of the forward voltage, so that this diode will attract a further increased fraction of the current, and so on. A pronounced negative temperature dependency with a V_F(T)-decrease of more than 2 mV/K endangers thermal instability in parallel connection. A diode like that in Fig. 5.11 (right) is, on the other hand, well suited for parallel connection.

Diodes connected in parallel are coupled thermally:

- via the substrate for paralleling of several diodes in a module,
- via the heat sink with paralleling of modules.

In the case of a weakly negative temperature coefficient, this coupling is usually sufficient to avoid a thermal runaway of the diode with the lowest forward voltage. In the case of diodes with a negative temperature coefficient of <-2 mV/K, it is recommended to decrease the current load in parallel connection to a lower value as would result from the sum of the current ratings of the single diodes. This measure is known as “derating”.

5.5 Relation Between Stored Charge and Forward Voltage

Especially with fast diodes, a trade-off between the demands for fast switching—low stored charge etc.—and for low forward voltage has to be made. If the carrier lifetime is adjusted to be low, then the forward voltage increases according to Eq. (5.61). According to (5.37), the voltage V_{drift} and the stored charge Q_F are related by the equation

$$Q_F = \frac{w_B^2 \cdot I_F}{V_{drift}\left(\mu_n + \mu_p\right)} \tag{5.64}$$

The forward voltage has been partitioned into the parts V_{drift} in the base region and the voltage drops $V_j(p^+n)$ at the pn- and $V_j(nn^+)$ at the nn^+-junction (see Eq. 5.29). These terms are combined in Eq. (5.32) to the junction voltage V_j, $V_F = V_j + V_{drift}$. Replacing V_{drift}, (5.64) results in

$$Q_F = \frac{w_B^2 \cdot I_F}{\left(V_F - V_j\right)\left(\mu_n + \mu_p\right)} \tag{5.65}$$

Equation (5.65), like Eq. (5.35), is a good approximation for $w_B/L_A < 4$. This hyperbolic relation is shown in Fig. 5.11. In this figure, Q_F according to (5.65) is drawn for a fast 600 V diode with $w_B = 65$ µm. For comparison, the experimental results for Q_{RR} of a fast diode with this dimensioning are also shown. Q_{RR} distinguishes from Q_F by the amount of charge recombining during the measurement duration.

A hyperbolic relation as shown in Fig. 5.12 results for every technology; how far it can be shifted to lower values, however, is an evaluation criterion for the specific design. The base-width w_B contributes squared to Q_F as well as to V_F. Therefore, w_B must be kept as low as possible by taking into account all requirements to the diode.

Nevertheless, up to now nothing has been stated about the time-dependent waveform in which the stored charge emerges during the reverse-recovery procedure. This, however, is most significant and is to be discussed in the following paragraphs regarding the turn-off behavior. However, first the turn-on behavior will be investigated.

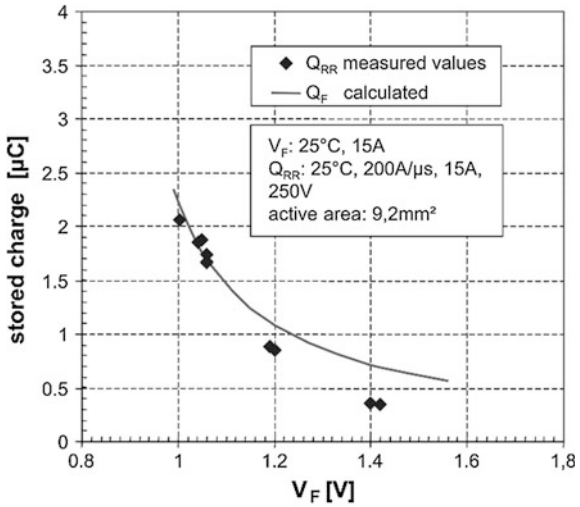


Fig. 5.12 Relation of stored charge to forward voltage for a fast 600-V diode

5.6 Turn-on Behavior of Power Diodes

At the transition of the diode into the conducting state, the voltage first increases to the turn-on voltage peak V_{FRM} (Forward-Recovery Maximum) before it drops down to the forward voltage. Figure 5.13 shows the definition of V_{FRM} and the turn-on time t_{fr}, in which t_{fr} is defined as the time interval between the instant of 10% forward voltage and the instant at which the voltage has dropped down again to the 1.1-fold value of the steady state forward voltage.

This old definition comes from a time in which thyristors were the dominating devices in power electronics; low current slopes were usual and V_{FRM} amounted to several volts. This definition does not pertain to freewheeling diodes and snubber diodes used in circuits with IGBTs as switching elements, because with them such steep slopes *di/dt* of the current occur that V_{FRM} may reach – for example in a poorly designed 1700 V diode – a value of 200–300 V which is more than 100-times the value of V_F. Reading out the time at the value of $1.1 \times V_F$ is no longer possible in the measurement.

A low V_{FRM} is one of the most important requirements for diodes in snubber- and clamping circuits, since these circuits work just after the diode has turned on.

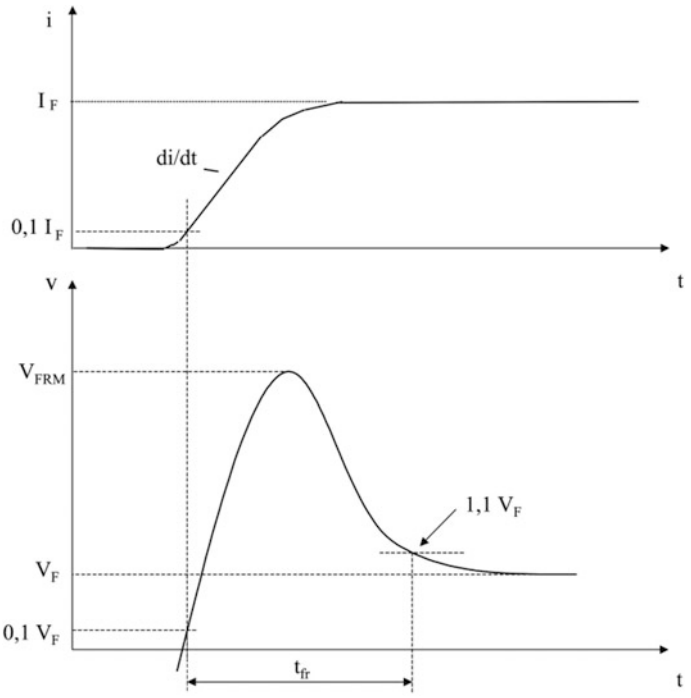


Fig. 5.13 Characteristic parameters of the turn-on behavior of power diodes

The forward-recovery voltage peak is an important feature also for freewheeling diodes designed for a blocking voltage of 1200 V and more. At turn-off of the IGBT, the freewheeling diode turns on; the *di/dt* at the IGBT turn-off creates at the parasitic inductance a voltage peak on which V_{FRM} is superimposed. The sum of both components can lead to a critical voltage peak.

The measurement of this behavior is not trivial, since the inductive component and V_{FRM} cannot be distinguished in an application-conform chopper circuit. The measurement is possible only at an open setup directly at the bond wires of the diode. Such a measurement is depicted in Fig. 5.14. Here, the turn-on of two diodes is shown, one of which (standard diode) is designed with a very wide w_B in order to achieve a soft recovery behavior at turn-off. For the CAL-diode presented in comparison, w_B is kept as low as possible. Under the same measurement conditions, i.e. the same parameters for controlling the IGBT turn-off, a V_{FRM} of 84 and 224 V results for the CAL-diode and the standard diode, respectively. It is possible to analyze the worst-case scenario. At a step function current form ($di/dt = \infty$), the maximally occurring voltage corresponds to the resistance of the base without carrier injection, multiplied by the current density:

$$V_{FRM} = \frac{w_B \cdot j}{q \cdot \mu_n \cdot N_D} \tag{5.66}$$

This equation can be used as long as $j < q \cdot N_D \cdot v_{sat}$. The design of a diode for higher blocking voltage requires a lower doping N_D and a wider base-width w_B, which increases the resulting voltage peak strongly. Figure 5.15 shows the results according to Eq. (5.66), whereby N_D and w_B were selected for fast diodes of different voltage ranges, and the mobility μ_n for $T = 400$ K from Fig. 2.12 was used.

For a diode designed for 600 V, a voltage peak amounting to only some 10 V can occur. For a diode designed for 1700 V, however, this peak may amount to more than 200 V, and for a diode designed for the voltage range of > 3000 V, more than 1000 V are possible. Moreover, the effects of recombination centers have not yet been considered in this calculation. It is known that the recombination center

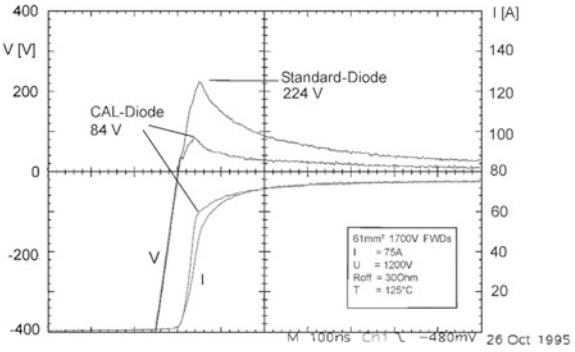


Fig. 5.14 Turn on of two diodes with different width w_B of the low-doped layer

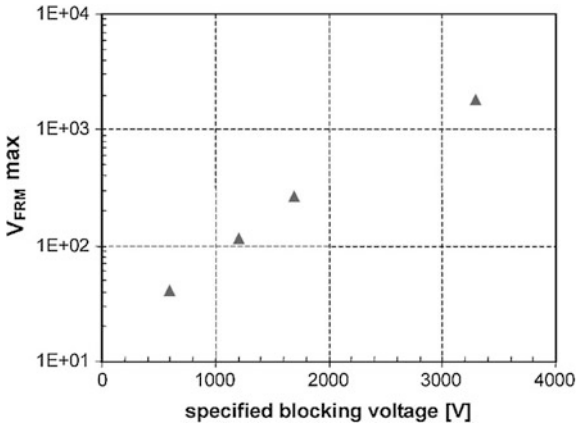


Fig. 5.15 Calculated worst-case voltage peak at turn-on of diodes at T = 125 °C. Compensation effects are not considered

gold has a compensating effect. In a low-doped n-region, the density of these acceptors compensates a part of the background doping density. The resulting decreased effective doping has to be applied in (5.66), and the voltage peak can then become significantly higher. In practice, there are no current slopes occurring with the form of a step-function, however, current slopes with di/dt in the range of > 2000 A/µs are to be expected in IGBT applications as shown in Fig. 5.14.

The importance of the turn-on behavior of freewheeling diodes was underestimated for a long time. Just after high-voltage IGBTs of > 3000 V were introduced and failures in the application occurred, the anti-parallel diode to the IGBT was found as a reason: if it creates a high forward-recovery peak V_{FRM}, this voltage peak is applied to the IGBT in the reverse direction. The reverse blocking capability is not specified for common IGBTs, since the collector-side pn-junction has no defined junction termination. To counteract this problem, more attention was paid to the turn-on of the freewheeling diodes.

Regarding switching losses, the turn-on behavior of the diode is not significant. Even if high voltage peaks occur, the turn-on process is very fast, and turn-on losses amount to only some percent of the turn-off losses or of the conduction losses of the diode. For thermal calculations, turn-on losses can be neglected in very most cases.

5.7 Reverse-Recovery of Power Diodes

5.7.1 Definitions

With the transition from the conducting to the blocking state, the charge stored in a diode has to be removed. This charge causes a current flow in the reverse direction of the diode. The reverse-recovery behavior signifies the time-dependent waveforms of this current and the corresponding voltage.

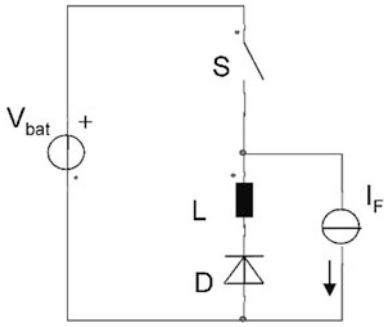


Fig. 5.16 Circuit for characterizing reverse-recovery behavior

The simplest circuit to measure this effect is the circuit according to Fig. 5.16, whereby S represents an ideal switch, I_F an ideal current source, V_{bat} an ideal voltage source, L an inductor and D the diode being considered. After closing the switch S, the progression of current and voltage as shown in Fig. 5.17 occurs at a the diode. Figure 5.17 exemplifies a diode with soft recovery behavior, whereas Fig. 5.18 depicts two examples of the current waveform of diodes with a snappy reverse-recovery behavior.

First, the definitions will be explained using the circuit in Fig. 5.16 and the waveform in Fig. 5.17. In the circuit of Fig. 5.16, after closing the switch S it holds

$$L \cdot \frac{di}{dt} + v(t) = -V_{bat} \tag{5.67}$$

where $v(t)$ is the time dependent voltage at the diode. First, during the current decay, the voltage at the diode is in the range of the forward voltage V_F which is in

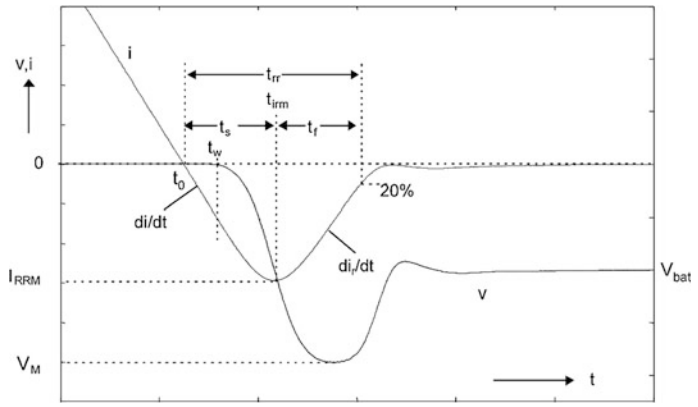


Fig. 5.17 Waveforms of current and voltage for a soft recovery diode during the reverse-recovery process in a circuit according to Fig. 5.16 and definitions of some characteristic values of the recovery behavior

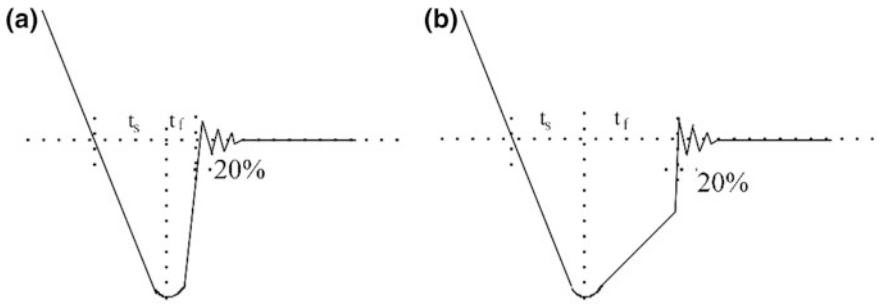


Fig. 5.18 Current waveform for two different possibilities of snappy reverse-recovery behavior

the range of 1 – 2 V, and $v(t)$ can be neglected. The current slope at commutation is determined by the voltage and inductance:

$$-\frac{di}{dt} = \frac{V_{bat}}{L} \tag{5.68}$$

The zero crossing point of the current occurs at t_0. At t_w the diode starts to take over the voltage; at this instant the pn-junction of the diode is free of charge carriers. At the same point in time, the current deviates from the linear slope. At t_{irm} the reverse current attains its maximum I_{RRM}. At t_{irm}, it holds that $di/dt = 0$ and from Eq. (5.67), $v(t) = -V_{bat}$ results.

After t_{irm} the reverse current decays down to the level of the static leakage current. The shape during this interval solely depends on the diode. If this decay is steep, a snappy reverse-recovery behavior is given. If this decay occurs slowly, however, a soft recovery behavior is indicated. This slope di_r/dt, which is often not linear, leads to an induced voltage $L \cdot di_r/dt$ that adds on the battery voltage.

The switching time t_{rr} is defined as the time between t_0 and the point in time at which the current has decayed down to the value of 20% of I_{RRM}. With the subdivision of t_{rr} in t_f and t_s as is shown in Fig. 5.17, formerly the following "soft factor" s was defined as the quantitative parameter of the reverse-recovery behavior:

$$s = \frac{t_f}{t_s} \tag{5.69}$$

whereby e.g. $s > 0.8$ signified that a diode can be called to be "soft".

This definition, however, is very insufficient. According to it, a current shape as in Fig. 5.18a would be snappy, but a current shape as in Fig. 5.18b would be accepted as soft. While in Fig. 5.18b $t_f > t_s$ is given and $s > 1$ according to (5.69), a very steep slope, a reverse current snap-off, occurs in a part of the reverse-recovery waveform.

The following definition of the soft factor is better:

$$s = \left| \frac{-\frac{di}{dt}\Big|_{i=0}}{\left(\frac{di_r}{dt}\right)_{\max}} \right| \tag{5.70}$$

The applied current slope must be measured at the zero crossing point, and the di_r/dt caused by the diode is measured at its maximal value. The measurement must be executed at less than 10% and at 200% of the specified rated current. For soft recovery the value of $s > 0.8$ is again required. With this definition, also a behavior like that shown in Fig. 5.18b is considered to be snappy. Additionally, this definition includes the observation that small currents are especially critical for reverse-recovery behavior.

The term di_r/dt determines the occurring voltage peak, and $v(t)$ in (5.67) has the maximal amplitude at the maximal slope:

$$V_M = -V_{bat} - L \cdot \left(\frac{di_r}{dt}\right)_{\max} \tag{5.71}$$

Thus, the voltage peak V_M occurring under special conditions or the induced voltage $V_{ind} = V_m - V_{bat}$ can be used as a quantitative definition for the reverse-recovery behavior. As conditions, V_{bat} and the applied di/dt must be indicated.

This definition, however, is also insufficient, because even more parameters have an influence on the reverse-recovery behavior:

1. The temperature: in most cases, high temperatures are more critical for the reverse-recovery behavior. For some fast diodes, however, room temperature or a lower temperature is a more critical condition for possible occurrence of snappy recovery behavior.
2. The applied voltage V_{bat}: higher voltage leads to worse recovery behavior.
3. The value of the inductor L: according to (5.71), with increased L the voltage at the diode is increased; this makes the conditions for the diode harder.
4. The commutation velocity di/dt: A rise in di/dt leads to greater danger of oscillations and snap-off of the current. The reverse-recovery behavior has an increased tendency towards snappy behavior.

All these different influences cannot be covered by a simple quantitative definition. The circuit according to Fig. 5.16 and the definitions according to Eqs. (5.69) and (5.70) can only be used to show the effect of different design parameters. In fact, the reverse-recovery behavior has to be evaluated by using the waveforms of current and voltage, which are measured under application-conform conditions.

The application-conform double-pulse measurement circuit is shown in Fig. 5.19. Compared to the circuit in Fig. 5.16, the ideal switch is replaced by a real switch, e.g. an IGBT. The ideal current source is replaced by an ohmic-inductive load consisting of R and L. The commutation velocity is given by the transistor; for

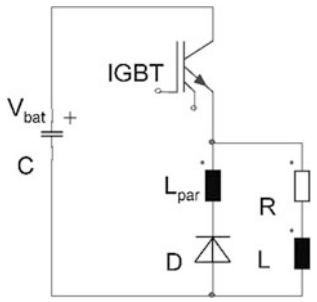


Fig. 5.19 Application-conform double-pulse circuit for measuring the reverse-recovery behavior

an IGBT it is adjustable by the resistor R_{on} in the gate circuit as described later in Chap. 10. V_{bat} is the battery voltage, which is assisted by a capacitor C. The wiring between capacitor, IGBT, and the diode together form a parasitic inductance.

In Fig. 5.20 the drive signals for the IGBT, the current in the IGBT and the current in the diode are shown for the double-pulse mode. By turning off the IGBT, the load current is transferred to the freewheeling diode. At the next turn-on of the IGBT, the diode is commutated – the point in time when the characteristic reverse-recovery of the diode occurs. At turn-on, the IGBT has to conduct additionally the reverse current of the freewheeling diode.

This reverse-recovery event is shown at higher time resolution in Fig. 5.21 for a soft recovery diode. Figure 5.21a presents the current- and voltage waveform in the IGBT and the resulting power loss $v(t)\cdot i(t)$ at the turn-on. In addition, Fig. 5.21b

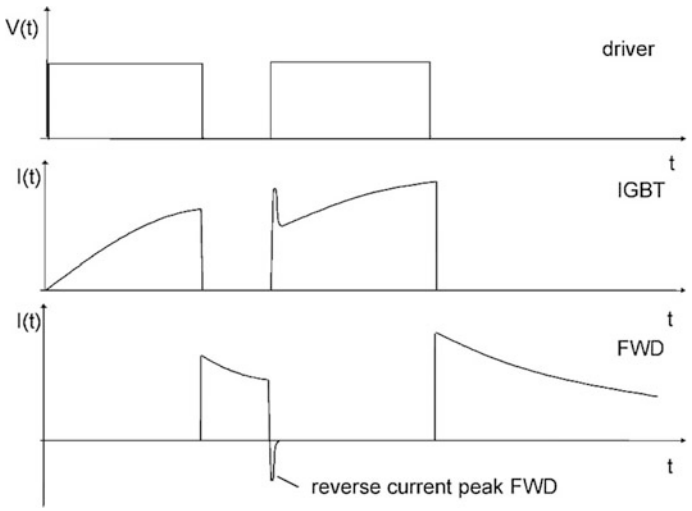


Fig. 5.20 Signal of the driver, current in the IGBT and in the freewheeling diode (FWD) at double-pulse measurement with the circuit according to Fig. 5.19

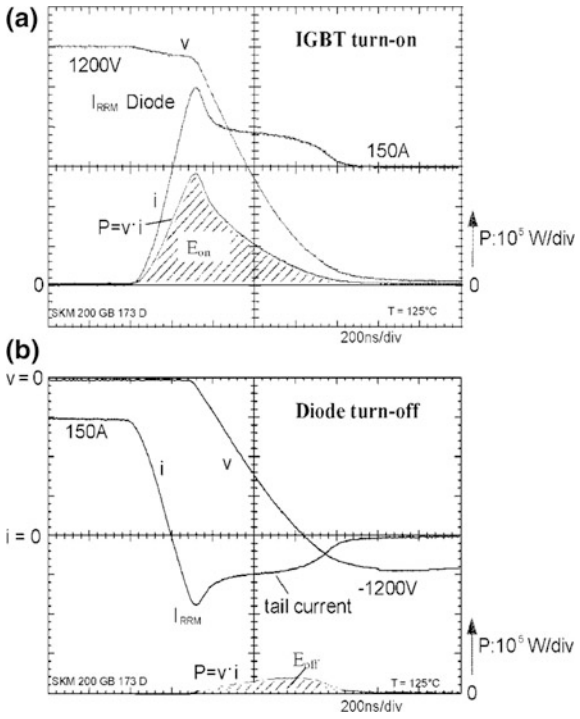


Fig. 5.21 Current waveform, voltage waveform and power losses at turn-on of the IGBT (**a**) and simultaneous turn-off of the diode (**b**) during the measurement of the diode recovery behavior in a double-pulse circuit according to Fig. 5.19

displays the current- and voltage waveforms for the freewheeling diode as well as the power losses in the diode.

While the IGBT has to conduct the reverse-recovery current maximum I_{RRM} of the freewheeling diode additionally to the load current, the voltage at the IGBT is still in the range of the battery voltage V_{bat} (1200 V in Fig. 5.21a). In this instant the maximum of the turn-on loss in the IGBT occurs.

The reverse-recovery current waveform of the diode can be divided into two phases:

1. The waveform until I_{RRM} and the subsequent decay of the reverse current with di_r/dt. In a soft recovery diode, $|di_r/dt|$ is in the range of $|di/dt|$. The reverse current peak I_{RRM} causes the most stress for the switching device.
2. The tail current phase during which the reverse current slowly phases out. A useful definition of the switching time t_{rr} is hardly possible for such a waveform. The tail current phase causes the main losses in the diode, since now there is a high voltage across the diode. Even though a snappy diode without tail current would feature less switching losses in the diode, it is detrimental for the application due to generated voltage peaks and oscillations. Slow and soft waveforms are desired. For the IGBT the tail current phase of the diode causes less stress, since in this phase the voltage at the IGBT has already decayed down to a low value.

The diode switching losses in Fig. 5.21b are represented on the same scale as those for the IGBT in Fig. 5.21a; the diode losses in the application are small compared to the losses in the IGBT. With regard to the total losses in the interaction of both devices, it is therefore important to keep the reverse-recovery current peak I_{RRM} low, and to take care that the main part of the stored charge of the diode is extracted in the tail phase. As the tail current causes the main part of the switching losses in the diode, it must also be limited. Typically, the switching losses in the diode are lower than those in the transistor (compare Fig. 5.21a, b). Regarding its contribution to the total losses, the most important characteristic for the diode is the reverse-recovery current peak I_{RRM} that must be as low as possible.

For a typical application in a current range of 100 A, in which the semiconductor devices of the chopper circuit are packaged inside of a single module, the parasitic inductance L_{par} is in the range of 40 nH or below. This leads to no significant overvoltage. Since there is no longer an ideal switch but rather a real transistor, the transistor still takes a part of the voltage during the reverse-recovery phase of the diode and lowers the applied battery voltage by $v_C(t)$. After turn-on of the transistor, (5.67) is now valid in a modified form:

$$L_{par} \cdot \frac{di}{dt} + v(t) = -V_{bat} + v_C(t) \tag{5.72}$$

The voltage occurring at the diode after I_{RRM} is now

$$v(t) = -V_{bat} - L_{par} \cdot \frac{di_r}{dt} + v_C(t) \tag{5.73}$$

where $v_C(t)$ is the voltage drop across the transistor in this phase. For soft recovery diodes it is typical that at moderate commutation velocities *di/dt* of up to 1500 A/µs and minimized parasitic inductance, the absolute value of the voltage at the diode $v(t)$ is smaller than V_{bat} and no voltage peak occurs.

As long as the circuit according to Fig. 5.19 is applied and the parasitic inductance is kept low, one can use the following definition:

A diode exhibits a soft recovery behavior if, under all relevant application conditions in an application conform circuit, no overvoltage is caused at the diode by a reverse-recovery current snap-off.

The relevant conditions cover the whole current range, all commutation velocities which can occur in the application and the temperature range from −50 °C up to 150 °C.

This definition is valid provided that there are not excessively high commutation velocities (> 6 kA/µs) or high parasitic inductances (> 50 nH) in the circuit. If L_{par} is increased and the switching characteristics of the IGBT approaches the characteristics of an ideal switch, which means that $v_C(t)$ approaches zero, then the circuit in Fig. 5.19 approaches the circuit in Fig. 5.16. In this case, voltages peaks are unavoidable also for soft recovery diodes.

In high-power modules with a rated current of 1200 A and more, 24 or more IGBT chips are connected in parallel. Such modules exhibit a large volume and it is very difficult to achieve a low parasitic inductance when connecting these modules to a three-phase inverter. In this range it makes sense to investigate the diode with an increased parasitic inductance. Thus, it should be investigated whether there are conditions at which a reverse current snap-off occurs. In such a reverse current snap-off, the waveform usually is similar to the waveform shown in Fig. 5.18b. The reverse current snap-off may even occur relatively late in the reverse recovery waveform, at the end of the tail current.

5.7.2 Reverse-Recovery Related Power Losses

The turn-off energy of the diode per switching event is generally (see Fig. 5.21b):

$$E_{off} = \int_{t_s + t_f} v(t) \cdot i(t) dt \tag{5.74}$$

A simplified estimation can be given for two cases. The first is the case of the circuit according to Fig. 5.16 and the waveform in Fig. 5.17. This waveform is drawn in a simplified way in Fig. 5.22.

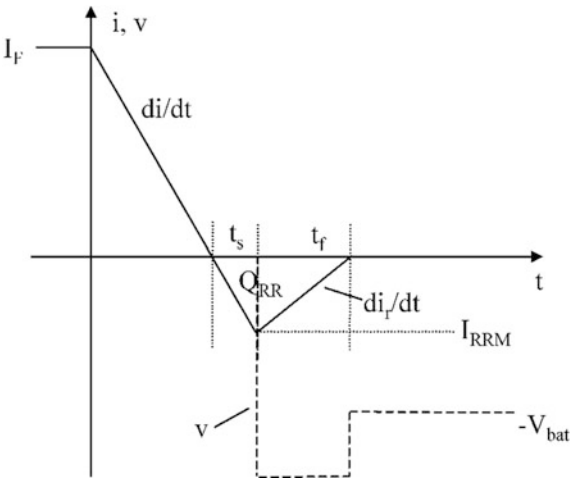


Fig. 5.22 Simplified waveform of current and voltage at the diode during turn-off in the circuit according to Fig. 5.16

During the time until current zero-crossing and during the time t_s, the voltage is simply assumed to be $v = 0$. For $t > t_s$ the diode takes over the voltage. If a linear decay of i_r during t_f is assumed, it holds during t_f that

$$i_r(t) = -I_{RRM} + \frac{I_{RRM}}{t_f} \cdot t \tag{5.75}$$

$$v = -V_{bat} - L \cdot \frac{di_r}{dt} = -V_{bat} - L \cdot \frac{I_{RRM}}{t_f} = const. \tag{5.76}$$

Thus, it follows:

$$E_{off} = \frac{1}{2} \cdot L \cdot I_{RRM}^2 + \frac{1}{2} \cdot V_{bat} \cdot I_{RRM} \cdot t_f \tag{5.77}$$

The first term on the right-hand side of (5.77) can be modified using Eq. (5.67). Accordingly, it holds that $L = -\frac{V_{bat}}{di/dt} = \frac{V_{bat}}{I_{RRM}/t_s}$. With this, one obtains from (5.77):

$$\begin{aligned} E_{off} &= \frac{1}{2} \cdot V_{bat} \cdot I_{RRM} \cdot t_s + \frac{1}{2} \cdot V_{bat} \cdot I_{RRM} \cdot t_f \\ &= \frac{1}{2} \cdot I_{RRM} \cdot t_{rr} \cdot V_{bat} = Q_{RR} \cdot V_{bat} \end{aligned} \tag{5.78}$$

The diode turn-off losses are thus directly proportional to Q_{RR}.

This simplified consideration is valid only for the circuit in Fig. 5.16 in which main effects are determined by the inductor L. Also for the second case, the application-conform circuit according to Fig. 5.19 with the waveforms according to Fig. 5.21, a simplified estimation of the switching losses is possible. The waveforms from Fig. 5.21 are illustrated in a simplified way in Fig. 5.23. Here, the parasitic inductance L_{par} is neglected, and the voltage slope at the diode is only determined by the voltage slope at the transistor $v_C(t)$. The voltage fall time of the transistor t_{fv} is assumed to be equal to the reverse current fall time t_f of the diode. The current- and voltage waveforms are again idealized by straight lines.

The reverse-recovery charge Q_{RR} of the diode is subdivided into the charge Q_{RS} occurring during the storage time t_s and the charge Q_{RF} occurring during the reverse current fall time t_f. It holds that $Q_{RR} = Q_{RS} + Q_{RF}$. The turn-off energy loss in the diode per switching event is then

$$E_{off} = \frac{1}{3} Q_{RF} \cdot V_{bat} \tag{5.79}$$

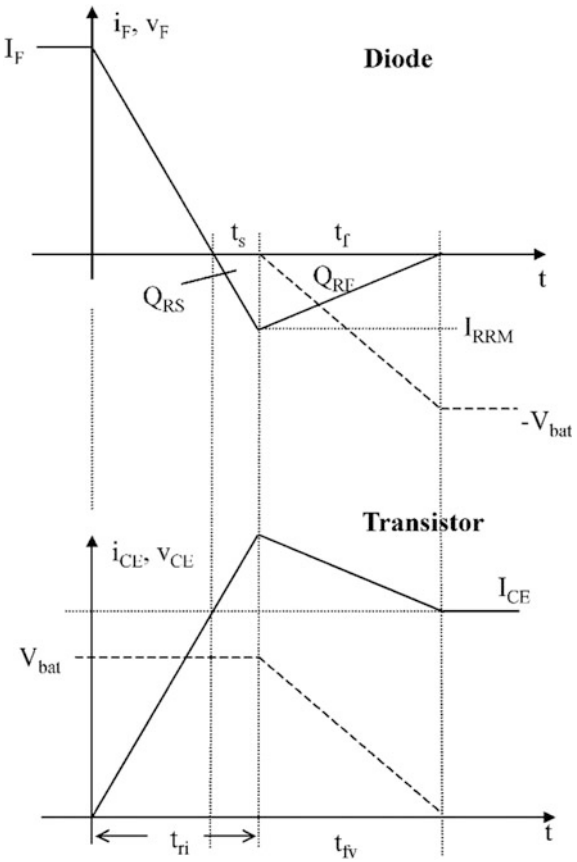


Fig. 5.23 Simplified waveforms of current and voltage at the diode and in the transistor for the circuit according to Fig. 5.19

The parameters Q_{RR} and I_{RRM} at specified conditions di/dt und V_{bat} are given in the data sheets of well specified, modern freewheeling diodes. From I_{RRM} and di_F/dt, one can calculate Q_{RS} as

$$Q_{RS} = \frac{1}{2} t_s \cdot I_{RRM} = \frac{1}{2} \cdot \frac{I_{RRM}}{di_F/dt} \cdot I_{RRM} = \frac{1}{2} \cdot \frac{I_{RRM}^2}{di/dt} \tag{5.80}$$

and because of $Q_{RF} = Q_{RR} - Q_{RS}$, it follows:

$$E_{off} = \frac{1}{3} V_{bat} \cdot \left(Q_{RR} - \frac{1}{2} \cdot \frac{I_{RRM}^2}{di/dt} \right) \tag{5.81}$$

If this result is compared with the result (5.78), the case of the inductor-determined turn-off process, then one can see: the turn-off energy loss is less than half of the estimation in Eq. (5.78). The transistor has relieved the stress for the diode by its $v_C(t)$ which drops across the transistor during the diode turn-off event and which slowly decays during the turn-on of the transistor.

Nevertheless, for this reduction of losses in the diode, the switching transistor has to pay the price during its turn-on. From the same simplified consideration in Fig. 5.23, it can be derived that assuming a freewheeling diode without reverse-recovery current maximum – and thereby without a stored charge – the idealized turn-on energy loss in the transistor would be

$$E_{on}(tr, id) = \frac{1}{2} \cdot (t_{ri} - t_s) \cdot I_{CE} \cdot V_{bat} + \frac{1}{2} \cdot t_{fv} \cdot I_{CE} \cdot V_{bat} \tag{5.82}$$

By the freewheeling diode, the following terms are generated additionally:

the dissipated energy because the current-increase time t_{ri} is prolonged by t_s; in this interval, the current as well as the voltage are high
the power losses caused by Q_{RS}
the power losses caused by Q_{RF}

This leads to additional losses ΔE_{on}—in the order of the enumeration:

$$\Delta E_{on} = t_s \cdot I_F \cdot V_{bat} + Q_{RS} \cdot V_{bat} + \frac{2}{3} Q_{RF} \cdot V_{bat} \tag{5.83}$$

Comparing this with Eq. (5.79), one can state that the losses in the transistor, generated by the diode, are higher than the losses in the diode itself by the first terms in Eq. (5.83) and additional twice of the diode losses according to Eq. (5.79) occurs as last term in Eq. (5.83).

The turn-on losses in the transistor amount to $E_{on}(tr) = E_{on}(tr,id) + \Delta E_{on}$. The sum of the losses caused by the diode in the transistor and in the diode is:

$$E_{off} + \Delta E_{on} = t_s \cdot I_F \cdot V_{bat} + Q_{RR} \cdot V_{bat} \tag{5.84}$$

This is significantly higher than the estimation in Eq. (5.78); in summary, an excessive price was paid for relieving the stress in the diode!

These estimations show that the requirement for diodes is a low reverse-recovery peak I_{RRM}, and thereby a t_s which is as low as possible.

In real circuits, a parasitic inductance that causes additional losses in the diode has to be considered. If the parasitic inductance dominates and if the relief of the diode by the voltage decay of the transistor can be neglected, then the losses in the diode again approach the situation that was described with Eq. (5.78). Comparatively high parasitic inductances have to be taken into account for example in some traction applications.

In the application, neither an ideal switch nor a circuit without any inductance can be presumed. Thus, Eq. (5.74) must be used for an exact determination of the switching losses. The waveforms of i and v are recorded with an oscilloscope, multiplied and integrated over the total switching time. The simplified estimation given here comply with measured values in a low-inductive circuit with an accuracy of ±20%.

5.7.3 Reverse Recovery: Charge Dynamic in the Diode

Figure 5.6 has shown the flooding of the base of the diode with free carriers for the forward conduction state. The reverse recovery behavior is determined by the internal behavior of the stored plasma and the time dependent shape during its removal. First, this shall be investigated qualitatively. Figure 5.24 depicts the simulation of the stored carriers in a snappy diode, whereas Fig. 5.25 shows the same for a soft-recovery diode.

At forward conduction, the n^--base of the diode is flooded with free electrons and holes in a range of 10^{16} cm^{-3}; they build up a neutral plasma in which the density of electrons n and of holes p are approximately equal. After commutation a neutral plasma zone with $n \approx p$ is still present in the diode until the time t_4. The removal of free carriers takes place towards the cathode by the electron current and towards the anode by the hole current, the plasma removal process occurs in the

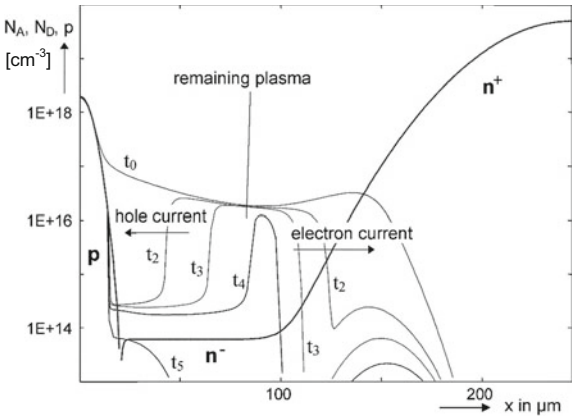


Fig. 5.24 Doping profile and concentration of the remaining plasma during reverse recovery in a snappy diode (ADIOS-Simulation)

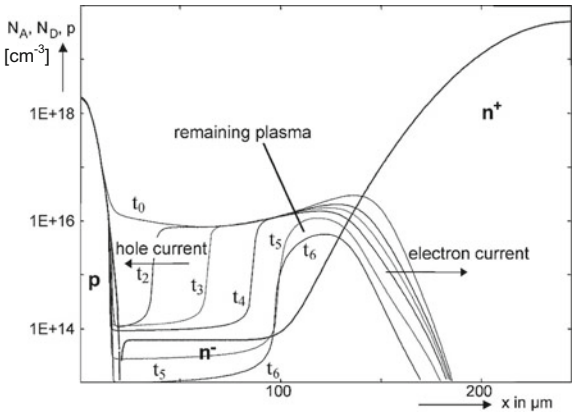


Fig. 5.25 Doping profile and concentration of the remaining plasma in a soft recovery diode (ADIOS-Simulation). Compared to Fig. 5.24 the difference is caused by an inhomogeneous carrier lifetime

outer circuit as reverse current. In the case of the snappy diode (Fig. 5.24), both edges of the plasma meet shortly after t_4, and suddenly the source for feeding the reverse current vanishes. The reverse current is interrupted abruptly, the reverse recovery behavior is snappy.

The time-dependent shape of the remaining plasma for a soft recovery diode is shown in Fig. 5.25. The doping profile is the same as in Fig. 5.24 in this example, but the on-state carrier concentration ($t = t_0$) is changed resulting from an inhomogeneous carrier lifetime, i.e. a low lifetime at the pn-junction, and a higher lifetime at the nn^+-junction. During the whole reverse recovery process, a neutral plasma remains inside the diode and feeds the reverse current. At the instant t_5 the diode has taken over the applied voltage. A plasma decay as shown in Fig. 5.25 leads to a tail current as depicted in Fig. 5.21b.

Whether soft recovery behavior is achieved depends on the time-dependent decay of the plasma and how this process is controlled suitably. It took a comparatively long time until the reverse recovery behavior was mastered.

For analyzing the dynamic behavior of the plasma, two cases will be investigated in more detail in the following. The first case refers to the effects during the increase in voltage and is based on the model of Benda and Spenke [Ben67]. With this model is investigated under which conditions a reverse current snap-off or a soft recovery behavior is to be expected. Thereafter, a further analysis is conducted for the case that the device has already taken the voltage with a soft-recovery shape of the current and that despite this, a snap-off does occur at the end of the tail current.

For investigating the first case based on the model of Benda and Spenke, the following simplifications are assumed:

- The pn- and nn^+-junctions are assumed to be abrupt, and the charge within the highly doped regions is neglected.
- The plasma edges are treated as abrupt.
- Since the amount of charge carriers removed by the current is much higher than the recombination of charge carriers, recombination in this time interval is neglected.
- Additionally, the hyperbolic asymmetric shape of the initial plasma (Fig. 5.6 respectively Eq. (5.27)) is, as a first step, simplified by a constant plasma density, whereby $n = p = \overline{n}$ is assumed.

Figure 5.26 illustrates this simplified model. In the plasma the current consists of electron current and hole current; it holds that $j = j_n + j_p$ whereby

$$\begin{aligned} j_p &= \frac{\mu_p}{\mu_n + \mu_p} j \\ j_n &= \frac{\mu_n}{\mu_n + \mu_p} j \end{aligned} \tag{5.85}$$

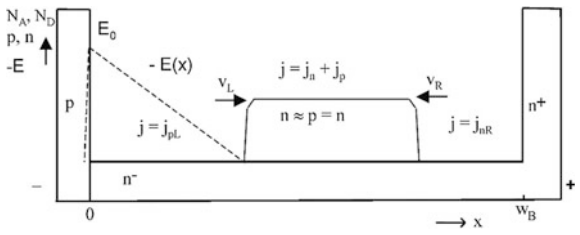


Fig. 5.26 Simplified drawing of the removal of the internal plasma in a diode

On the left side of the plasma zone, towards the pn-junction, a part of the n^--base has become free of carriers and a space charge is building up. The current flows here as a hole current $j = j_{pL}$. From the continuity condition for the current at the border of plasma and space charge, it follows:

$$j_{pL} = j_n + j_p \tag{5.86}$$

For the difference of the hole current, it is therefore valid at this position that

$$j_{pL} - j_p = \Delta j_p = j_n \tag{5.87}$$

Using (5.85) it follows:

$$\Delta j_p = \frac{\mu_n}{\mu_n + \mu_p} j \tag{5.88}$$

Both sides of Eq. (5.85) multiplied by the time interval dt lead to a differential charge

$$\Delta j_p \cdot dt = \frac{\mu_n}{\mu_n + \mu_p} j \cdot dt \tag{5.89}$$

This differential charge corresponds to a charge $q \cdot \overline{n} \cdot dx$ stored in a volume element dx while the plasma zone is shortened by dx. Consequently, (5.89) becomes

$$q \cdot \overline{n} \cdot dx = \frac{\mu_n}{\mu_n + \mu_p} j \cdot dt \tag{5.90}$$

and for the velocity of the movement of the left-hand side border of the plasma to the right hand side, one obtains

$$|v_L| = \frac{dx}{dt} = \frac{\mu_n}{\mu_n + \mu_p} \cdot \frac{j}{q \cdot \overline{n}} \tag{5.91}$$

Analogously, this can be analyzed for the right side of the plasma zone. From the continuity condition for the current at the border between the plasma zone and the plasma-free zone at the right-hand side, it follows:

$$j_{nR} = j_n + j_p \tag{5.92}$$

and this leads to

$$|v_R| = \frac{\mu_p}{\mu_n + \mu_p} \cdot \frac{j}{q \cdot \overline{n}} \tag{5.93}$$

For silicon, with $\mu_n \approx 3\mu_p$, one obtains $v_L \approx 3v_R$: The plasma zone is removed from the side of the pn-junction thrice as fast as from the side of the nn^+-junction. With the assumed constant density of free carriers in the plasma, both plasma fronts will finally meet at

$$w_x = \frac{v_L}{v_L + v_R} \cdot w_B = \frac{3}{4} w_B \tag{5.94}$$

The electric field is built up by the space charge in the part of the n^--layer between the pn-junction and border of the plasma. The electric field cannot penetrate the plasma zone, since this is neutral. Hence, the electric field will have a triangular shape. The voltage across the device corresponds to the area below the line $-\mathsf{E}(x)$ in Fig. 5.26. The plasma-free zone at the right side does not contribute to the voltage in this first approximation.

Therefore, the device can take over as much voltage as possible with a width w_x before a reverse current snap-off occurs. This voltage shall be designated as threshold voltage for snappy behavior V_{sn}. Since electrons are not present in the space-charge region, one obtains

$$V_{sn} = \frac{1}{2} \frac{q \cdot N_D}{\varepsilon} w_x^2 \tag{5.95}$$

The current j in this region flows as a hole current, and the holes have the same polarity as the positively charged donor ions of the background doping. Equation (5.95) is valid as long as the density of holes p, which are traveling through the space charge, is low, $p \ll N_D$, and can be neglected. With increasing p, N_D has to be replaced by $N_{eff} = N_D + p$. Therefore, the gradient of the electric field and thereby the area below the line $-\mathsf{E}(x)$ is increased by the current. The device can thus take over more voltage. Maximally, this voltage can take the value at

which avalanche breakdown sets in. Rewriting Eq. (5.1), one obtains the maximal voltage V_{sn} as a function of w_x to be

$$V_{sn} = \left(\frac{w_x^6}{2^4 \cdot B}\right)^{\frac{1}{7}} \tag{5.96}$$

This is the maximal voltage for triangular electric field shape. Equation (5.96) predicts a higher value for V_{sn} than Eq. (5.95). In most cases, the prediction from Eq. (5.95) is too low, especially for switching with high *di/dt* which is typical in applications with IGBTs as switching devices; at these conditions the hole concentration p cannot be neglected. Equation (5.96) is closer to the experimental observation. It should be noted here that V_{sn} is always significantly lower than the breakdown voltage V_{BD} of the device. Since with a wide w_B also increases w_x, this shifts the snap-off of the reverse current to a higher voltage.

Earlier, some suggestions to achieve soft-recovery behavior in fast diodes have been based on this movement of the plasma fronts and proposed a wide w_B [Mou88]. Even some recent solutions are applying this approach. Here, the diode must be designed for triangular field shape (NPT), and w_x is determined using Eq. (5.1). Moreover, the doping is chosen as high as possible for the respective voltage according to Eq. (3.80), w_x results from (3.85) or (5.1) and finally, according to Eq. (5.94), $^1/_3 w_x$ is added to determine w_B.

As was already mentioned in Sect. 5.3 in the investigations on dimensioning of fast diodes, the minimal width of the n^--layer of a NPT diode is the $2^{2/3}$-fold of the minimal width of the n^--layer of a PT diode – the diode with the smallest width of the n^--region for the required voltage. Compared to the minimal width $w_{Bmin} = w_{B(PT,lim)}$ of Eq. (5.12), the suggestions above lead to

$$w_B = \frac{1}{0.63} \cdot \frac{4}{3} w_{B\,min} \cong \sim 2,1 \cdot w_{B\,min} \tag{5.97}$$

This results in a significant increase in the forward voltage drop, to which w_B contributes with the power of two or even exponentially. To avoid this high forward voltage, the charge-carrier lifetime can be increased, but this contradicts the requirement that the diode must be fast. Hence, further measures are necessary to achieve soft-recovery behavior, especially under condition of fast switching events in IGBT circuits. With the behavior of the internal plasma as in Fig. 5.25, a soft recovery is obtained without making w_B as thick.

Up to now, the plasma profile was assumed to be homogeneous and constant over the base of the diode, which contradicts reality. However, this simplified approach allows useful conclusions to be drawn also for an inhomogeneous distribution. According to Fig. 5.5 or Eq. (5.27), the plasma density is increased at the pn-junction because of the different mobilities of electrons and holes. This is now taken into account by the densities $\overline{n}_L$ at the side of the pn-junction and $\overline{n}_R$ at the

side of the nn$^+$-junction. The term $\overline{n}_L$ stands for the average density of carriers close to the pn-junction. Division of (5.93) with (5.91) leads to

$$\frac{v_R}{v_L} = \frac{\mu_p}{\mu_n} \cdot \frac{\overline{n}_L}{\overline{n}_R} \tag{5.98}$$

Furthermore, the quantity $\eta = p_L/p_R$ introduced Sect. 5.4.5 in Eq. (5.55) shall be used, and we approximate that this proportionality is valid within some distance to the respectively junction, e.g. $\eta = \overline{n}_L/\overline{n}_R = \overline{p}_L/\overline{p}_R$. Inserted into (5.94), this results in:

$$w_x = \frac{1}{1 + \frac{\mu_p}{\mu_n} \cdot \eta} \cdot w_B \tag{5.99}$$

If the profile in Fig. 5.6 is simply expressed with $\eta = 2$, then (5.99) yields

$$w_x = \frac{3}{5} \cdot w_B \tag{5.100}$$

The diode would snap-off even at a lower voltage than estimated with Eq. (5.94), or a diode with abrupt highly doped boarder regions and homogeneous lifetime in the base would have to be made even thicker than according to Eq. (5.97)!

On the other hand, if the profile in Fig. 5.6 can be inverted to maintain a higher density of carriers at the nn$^+$-junction compared to the pn-junction, this will be advantageous [Sco89]. If, for example $\eta = 1/3$, then Eq. (5.99) leads to $w_x = 0.9 \cdot w_B$, and w_B would have to be widened by a much smaller amount to obtain a sufficient w_x. Such a distribution can be achieved by special structures of the p-emitter, or by using a p-region which is much lower doped than the n$^+$-region, or by an inhomogeneous carrier lifetime which, at the pn-junction, is much lower than deep in the base. Such measures are used in modern fast diodes (see Sect. 5.7.4).

The above analytical approach using a simplified model gives us a good understanding of the physics of the recovery process. Considering the results of the numerical simulation shown in Fig. 5.24, one notices the following deviations from the simple model:

- The p and n$^+$ region usually have diffusion profiles; hence, the pn- and nn$^+$-junction are not abrupt.
- The very low gradient of the doping concentration at the nn$^+$-junction in the figure results in a later start of the carrier removal at the nn$^+$-junction than at the pn-junction. Therefore, the low doping gradient at the nn$^+$-junction is advantageous for the recovery behavior. The doping gradient at the pn-junction, on the other hand, should be as abrupt as possible, because this involves an early rise in the reverse voltage and a reduced peak reverse current.

- Of course, the gradients at the edges of the plasma cannot be infinite. However, the non-abrupt transition to the depletion regions does not greatly affect the velocity at the edges, because after a time interval dt the form of the carrier concentrations at the boundaries is approximately unchanged and only the thickness of the uniform plasma region is reduced.

Another aspect not included in the above model is avalanche generation which often takes place during fast switching, since free carriers, still present in the space-charge region, enhance the electric field. This "dynamic avalanche" will be discussed later in Chap. 13.

Now the second case will be considered: the device has successfully sustained the interval of the voltage increase, whereby the reverse recovery in this interval was soft. The device has taken over the applied voltage, but the plasma was not completely removed. Resulting from the remaining plasma, a tail current still flows. This tail current flows through the space charge as hole current, there holds $j = j_p$. Under the given condition of a high field, the holes flow with the drift velocity $v_{d(p)}$, which approximates the saturation drift velocity v_{sat}. Accordingly, the hole density in the space charge is

$$p = \frac{j}{q \cdot v_{sat}} \tag{5.101}$$

This influences the gradient of the electric field in the space charge region as

$$\frac{d\mathsf{E}}{dx} = \frac{q}{\varepsilon}(N_D + p) \tag{5.102}$$

This situation is illustrated in Fig. 5.27 for a diode with low base-doping N_D. The voltage can be assumed to be constant in the investigated time interval: consequently, the area under $-\mathsf{E}(x)$ is constant. The hole density is one factor in $d\mathsf{E}/dx$,

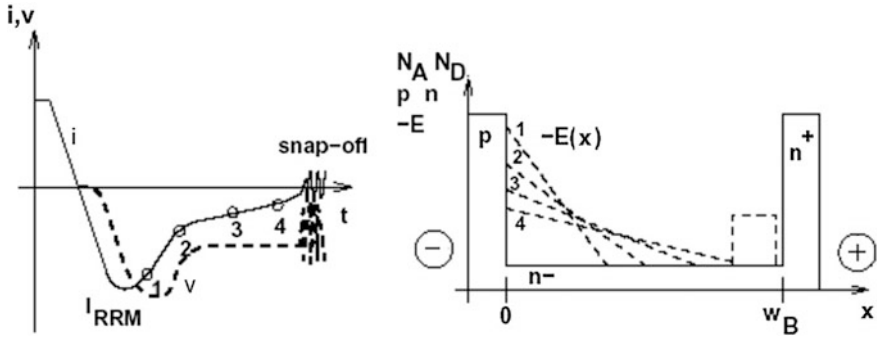


Fig. 5.27 Reverse current snap-off at the end of the tail current. Left: Waveform of current and voltage. Right: Electric field at different points in time

and p is determined by the hole current extracted from the remaining plasma. This hole current leads to a removal of the remaining plasma. By applying Eq. (5.101), (5.91) is changed to:

$$|v_l| = \frac{dx}{dt} = \frac{\mu_n}{\mu_n + \mu_p} \cdot \frac{p \cdot v_{sat}}{\overline{n}} \tag{5.103}$$

As p and j decrease, $d\mathsf{E}/dx$ becomes lower, and the space charge layer widens. However, if the space charge reaches the end of the base while still a significant current is flowing, the source of the current will suddenly vanish and the current will then snap-off. The electrical field springs from a triangular to a trapezoidal shape.

To avoid this effect, the device must be capable of taking the space charge at a given voltage without a punch of the space charge to the n^+-layer. The voltage limit, at which the space charge reaches the n^+-layer at a given background doping N_D and a base-width w_B, is:

$$V_{sn} = \frac{1}{2} \frac{q \cdot N_D}{\varepsilon} w_B^2 \tag{5.104}$$

For this case which is described in more detail in [Fel04], w_x can be set to w_B. As long as the battery voltage V_{bat} stays below V_{sn}, no reverse current snap-off will occur. In the static case or upon occurrence of voltage peaks, the diode can bear much more voltage, since then a trapezoidal space charge can be built up, and the breakdown voltage V_{BD} is much higher than V_{sn}.

The simplified description up to now can not give an answer how a diode transits from the plasma in Fig. 5.26 to that in Fig. 5.27, since only the field-induced drift components of the current were considered. Taking into account the diffusion components, the movement of the plasma layer backwards to the cathode can be described. The diffusion components on the right side of the plasma become dominating in the tail current phase, and the remaining plasma can move from the position in Fig. 5.26 to that in Fig. 5.27. This is explained in [Bab08].

The effect of reverse current snap-off in the tail phase can especially occur with diodes designed for higher voltages (> 2000 V). In such applications, often a low di/dt is applied, because the circuit contains significant parasitic inductances. This is the case in some applications for very high power control. Usually in such applications, the battery voltage is limited to 66% of the breakdown voltage V_{BD} for which the device is designed. In order to keep V_{sn} high, the doping N_D is not allowed to be too low. This, however, contradicts the demands of cosmic ray stability, see Sect. 12.8. Hence, an optimal trade-off has to be found.

5.7.4 Fast Diodes with Optimized Reverse-Recovery Behavior

Recombination centers are implemented in all fast silicon pin-diodes. The characteristics of the employed recombination centers gold, platinum and radiation induced centers have been described in Sects. 2.7 and 4.9. With the density of recombination centers, the charge carrier lifetime and thereby the stored charge Q_{RR} is decreased. However, there is no direct relationship between the concentration of recombination centers – as long as their axial distribution is kept constant – and the form of the reverse current decay. Therefore, it cannot be determined by the recombination center density whether the behavior will be soft or snappy.

It was already shown that the recovery behavior will become soft, if only the width w_B of the lowly doped base region is wide enough. Yet this leads to very high forward-conduction losses and/or switching losses, which can not be accepted in most cases. The modern concepts aim to adjust soft recovery behavior without using a strongly enhanced base-width.

5.7.4.1 Diodes with a Doping Step in the Low-Doped Layer

To avoid an excessively wide w_B and to minimize the disadvantages resulting from it, in 1981 an n^--layer with a step in the doping density was suggested by Wolley und Bevaqua [Wol81]. The doping profile of this diode is shown in Fig. 5.28. Approximately in the middle of the base, the doping is augmented by a factor of 5 – 10. Such layers are manufactured using a two-step epitaxy process.

When the space charge builds up and the field penetrates the higher doped layer n_2, it decreases there with a steeper gradient. At turn off, the remaining plasma zone is located in the layer n_2. The voltage which the device can take corresponds to the area below the line $-\mathsf{E}(x)$. This area is larger than that for a triangular field shape. The threshold voltage for possible snappy reverse-recovery behavior is shifted to a higher value. This measure is nowadays often used for diodes manufactured in epitaxial technology in the voltage range of up to 600 V. For diodes of higher voltage, especially above 1200 V, the fabrication of an epitaxial layer with the

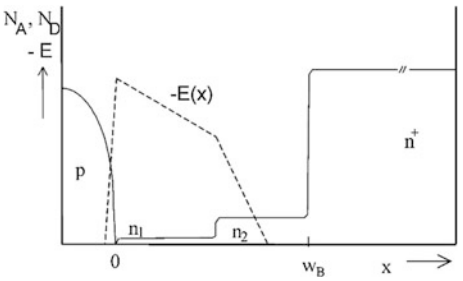


Fig. 5.28 Diodes with a step in the doping concentration of the low-doped layer

necessary thickness is too laborious. For soft-recovery behavior under all conditions relevant for the application, usually one of the following concepts is used.

5.7.4.2 Diodes with Anode Structures for Improving the Recovery Behavior

It was already shown that the carrier concentration in the flooded base of the pin-diode with highly doped boarder regions is higher at the pn-junction than at the nn^+-junction (see Fig. 5.6). This is disadvantageous for the reverse-recovery behavior. Therefore, concepts were developed to invert this distribution: the concentration shall be higher at the nn^+-junction than at the pn-junction. This principle was explained already in [Sco89], and there were several approaches to realize this by using structures in the p-anode layer.

For example, Schottky junctions cannot inject holes. Thus, it is obvious that implementation of Schottky junctions on a part of the area will lead to the desired distribution, considering the average concentration over the area. This measure is often discussed in the literature.

The "Merged Pin-Schottky" (MPS) diode consists of sequenced p-layers and Schottky regions [Bal98] (Fig. 5.29a). The distance between the p-layers is chosen to be so small that, in case of blocking voltage, the Schottky junction is shielded from the electric field, and only low field-strength occurs at its position. Consequently, the high leakage current of a Schottky junction is avoided. Figure 5.30 presents the forward characteristics of a MPS diode similar to that of [Bal98] calculated with the device simulator TCAD DESSIS [SYN07]. The structure with $w_B = 65$ µm is designed for a blocking voltage of 600 V. Compared here are the forward characteristics of the pin region, of the Schottky region and of the parallel connection of both in a MPS diode. At low current density the MPS diode approaches the characteristics of a Schottky diode. In the range of 200 A/cm^2, the typical current density of fast 600-V diodes at the rated current, the advantage caused by the Schottky regions is only small. At increased current density, the MPS diode has a higher forward voltage because of the loss of area of p-emitter regions.

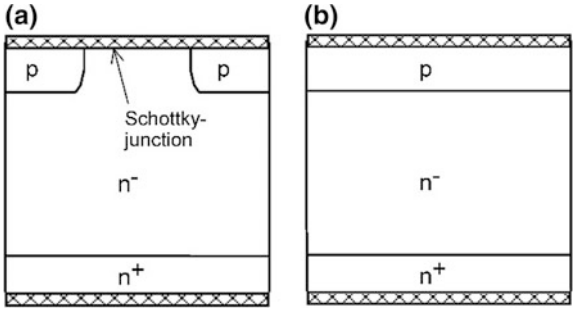


Fig. 5.29 p-emitter for improving the reverse-recovery behavior: **a** Emitter structures of the Merged Pin/Schottky diode **b** uniformly reduced p-doping, low emitter efficiency

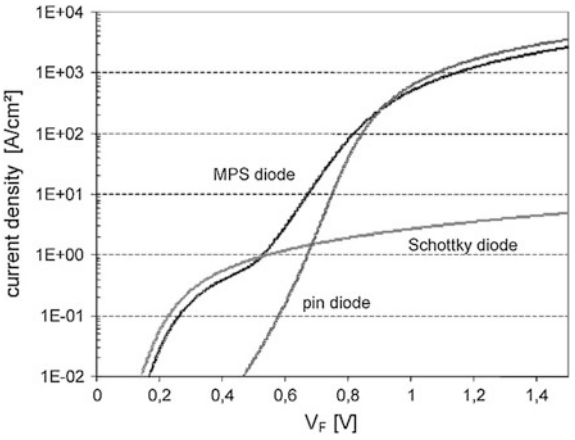


Fig. 5.30 Forward characteristics of a MPS diode with 50% Schlottky area. The characteristics of the pin-region and of the Schottky region are shown in comparison. Simulation using TCAD DESSIS

If the MPS diode is designed for a blocking voltage of 1000 V and greater, then the range of lower forward-voltage drop shifts to lower current density, because then the voltage drop in the low-doped base region predominates. But the effect, that the area of the p-region is reduced and thereby the injection of carriers at the anode side of the device is reduced, maintains its function. An inverted distribution of free carriers is formed (see Fig. 5.31, right-hand side).

Since the p-layers must be arranged closely to shield the Schottky regions from the electric field, the possibility is limited to make their share of the surface area small. A further idea to improve the MPS diode is the "Trench Oxide Pin Schottky" (TOPS) diode [Nem01] which was presented by Fuji. The structure is shown in Fig. 5.31. The p-anode regions are placed at the bottom of the trench cells. A Schottky contact is formed at the semiconductor surface. Hole injection takes place only by the p-layers that are connected via the resistor of the polysilicon layer in the trench. Thus, the integral hole injection over the area is strongly reduced.

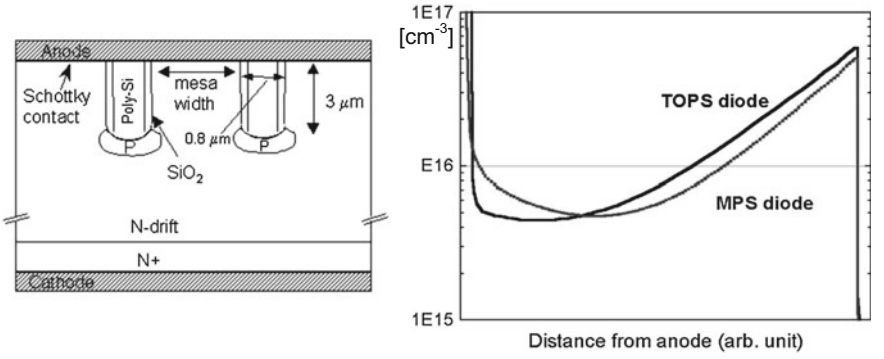


Fig. 5.31 Trench Oxide Pin Schottky (TOPS) diode. Structure (left) and vertical distribution of holes in the forward-conduction state (right). Figure taken from [Nem01] © 2001 IEEE

This results in a profile of free carriers as is shown in Fig. 5.31 on the right side, which exhibits a stronger inversion of the internal plasma.

A soft-recovery behavior can be expected from this plasma profile. The p-regions at the bottom of the trenches effectively shield the surface against the electric field; therefore, the leakage current is kept low with a narrow arrangement of the trenches.

There are several further concepts of structures at the anode side, among them, also structures with diffused p^+- and n^+-regions. What they all have in common is the reduction of the area of layers that are injecting holes and, thus, the reduction of the density of free carriers at the pn-junction.

5.7.4.3 The EMCON-Diode

Instead of reducing the emitter area by emitter structures, an homogeneous p-layer with high emitter recombination, as is shown in Fig. 5.29b, can also lead to the desired inverted distribution of the plasma. This concept is applied in the "Emitter Controlled" (EMCON) Diode [Las00]. It needs much less effort in production compared to the MPS diode or to the TOPS diode.

The EMCON diode utilizes a p-emitter of low emitter efficiency. The emitter parameter h_p, introduced in Eq. (3.98), can be expressed for a p-emitter which is not too highly doped as follows:

$$h_p = \frac{D_n}{p^+ \cdot L_n} \tag{5.105}$$

To reduce the emitter efficiency γ according to Eq. (3.100), h_p must have a high value. According to Eq. (5.105), this can be done if the doping density of the emitter p^+ is chosen low and also if the effective diffusion length L_n is adjusted to a low value. Both measures are used in the EMCON diode. p^+ must be sufficiently high to avoid a punch-through of the electric field to the semiconductor surface. For a thin p-layer, L_n is approximately the same as the penetration depth of the p-layer w_p, and also this depth is small in an EMCON diode. Under this condition it holds that

$$h_p = \frac{D_n}{p^+ \cdot w_p} = \frac{D_n}{G_n} \tag{5.106}$$

where $G_n = p^+ \cdot w_p$ is the Gummel-number of the emitter under the condition of an abrupt emitter [Sze81]. The term G_n represents the number of doping atoms per area. For the diffused emitter of the EMCON diode, it is more precisely expressed as:

$$G_n = \int_0^{w_p} p(x)dx \tag{5.107}$$

Using the Gummel-number according to (5.107) increases h_p compared to the abrupt emitter. High h_p and therewith low γ leads to a decreased p_L, as is necessary to create the inverted distribution. According to (5.52), h_p is the dominating factor for the emitter recombination. High h_p means that a significant share of the total recombination takes place inside the p-emitter or at the surface.

An emitter of high doping and high efficiency is applied at the cathode side of the EMCON diode; hence, the plasma density at this side is high.

Figure 5.32 depicts the turn-off waveform of an EMCON-HE diode. The measured turn-off behavior is compared to the numerical device simulation. Here, the simulated waveform agrees sufficiently well with the measured characteristic. The numerical simulation enables visualization of the effects inside the device. The simulated distribution of the density of free carriers is shown in Fig. 5.33 for the indicated time steps in Fig. 5.32.

At forward conduction the diode is flooded with free carriers. In Fig. 5.33 the rhombic dots show the measured carrier distribution in an EMCON diode at forward conduction as is obtained by the internal laser deflection method [Deb96]. The density of free holes calculated with the device simulator (line A in Fig. 5.33 for instant A in Fig. 5.32) is in good agreement with the measured density. The hole density represents the plasma density; $n \approx p$ applies for the regions of high flooding below the lines A, B, C etc. For the initial distribution at instant A, it holds that $\eta = p_L/p_R \approx 0.25$; this is attained by a strong emitter recombination in the anode.

During commutation and change of the polarity of the voltage (C, D, E in Fig. 5.32 and 5.33), the hole current flows to the negatively poled anode on the left side, and the electron current flows to the positively poled cathode. As is illustrated in Fig. 5.33, the removal of the stored charge occurs for the instants B, C, D, E. At the instant C, the diode has reached the maximal reverse current. After that, a current can still flow resulting from the removal of the still existing plasma which ensures a soft recovery behavior.

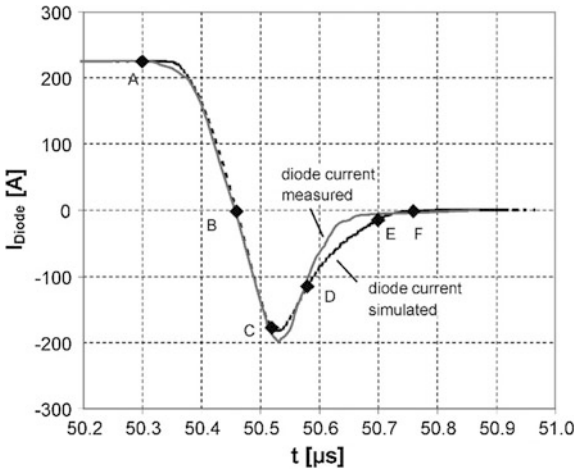


Fig. 5.32 Turn off behavior of a 1200 V EMCON-HE diode, measured and simulated at 600 V, 25 °C, 225 A/cm^2

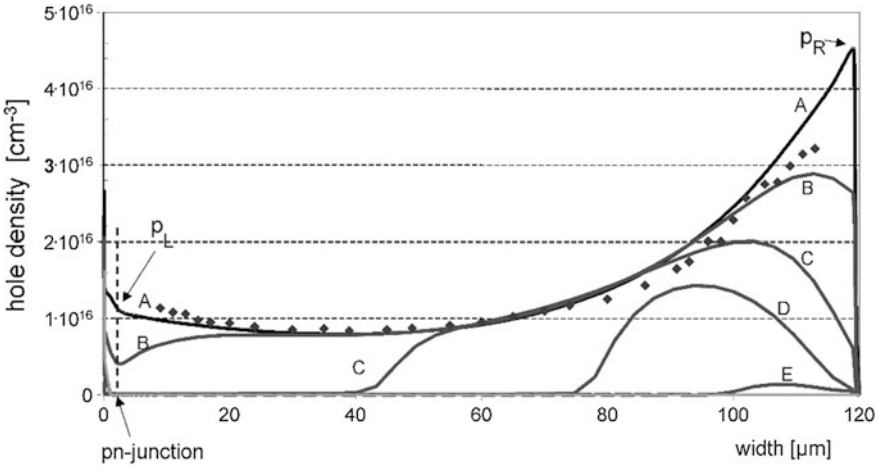


Fig. 5.33 Carrier distribution in an EMCON diode for forward conduction and during commutation. Measured plasma distribution in the forward-conduction state (rhombuses), simulated hole distribution for the forward-conduction state (line A), removal of the internal stored free-carriers (on example of holes) during commutation (lines B–E)

The low p-emitter efficiency of the EMCON diode leads to the disadvantage of a reduced surge current capability. A structure proposed in [Sco89], the so-called SPEED-diode containing high p-doped regions, removes this drawback to a mayor part.

5.7.4.4 The CAL-Diode

For the carrier lifetime adjustment in the diodes, often a platinum diffusion is used, whereby the vertical profile of the recombination centers (Fig. 4.28) results from the diffusion process and cannot be modified. The advantage of an adjusted profile of recombination centers, created by implantation of light ions, was already recognized early [Sil85, Won87]. But at that time this technology, which requires particle accelerators of energy of up to 10 meV, was only available for research issues. The situation changed in the early 90s of the last century. Then, the interest of basic researchers in high-energy physics migrated to the GeV region, and particle accelerators in the medium energy range became available for semiconductor production.

The first diode that reached the maturity of a series product with this technology was the so-called "Controlled Axial Lifetime" (CAL) diode [Lut94], designed for a blocking voltage of 1200 V in 1994. Meanwhile several manufacturers have applied this concept, whereby devices of up to 9 kV have been realized and are commercially available.

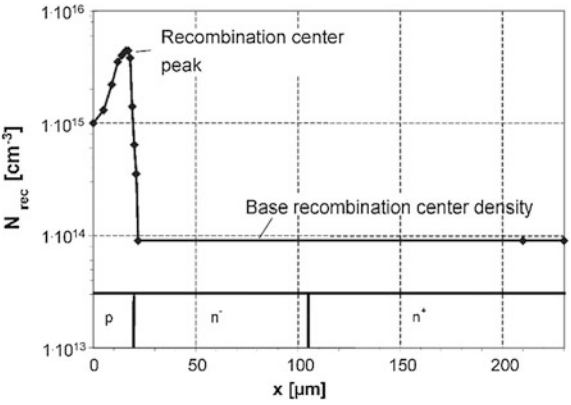


Fig. 5.34 Profile of recombination centers in the CAL diode (scheme)

Figure 5.34 illustrates the profile of recombination centers in the CAL diode. The recombination center peak is created by implanting He^{++}-ions. From this implantation, a profile of lattice defects similar to those in Fig. 4.19 results. Its depth can be adjusted by the energy of the helium-implantation, and its peak density by the dose. In combination with this implantation, the base recombination center density is adjusted as homogeneous across the base, preferably by electron irradiation. Three degrees of freedom – the recombination center peak depth, its height and the base recombination center density – are available for adjusting the reverse-recovery behavior.

It is most suitable to locate the recombination center peak close to the pn-junction. The main requirement is a low reverse-recovery current peak I_{RRM}. For this the pn-junction must be free of carriers at an early moment of the reverse recovery period. The relation between forward voltage V_F and I_{RRM} is better, the closer the recombination center peak is located to the pn-junction. In the CAL diode, the peak of radiation-induced recombination centers is located in the p-layer close to the pn-junction as is shown in Fig. 5.34. Consequently, the main part of multi-vacancies, which act as generation centers (see Sect. 4.9), is outside of the space charge, and the arrangement leads to a low leakage current.

An inverted plasma distribution in the on-state is achieved by this arrangement of the recombination center peak. The on-state plasma distribution shown in Fig. 5.25 is calculated with a recombination center profile according to Fig. 5.34.

The reverse-recovery behavior of the CAL diode was already displayed in Fig. 5.21b. The reverse recovery current peak, which can be adjusted by the height of the recombination center peak, is reduced, and the main part of the stored charge is extracted during the tail current phase. The tail current can be adjusted by the base recombination center density. An increased base recombination center density shortens the tail current time, but this has the drawback of increased forward voltage. With the given degrees of freedom, the reverse recovery behavior can be adjusted in a wide range. Thus, a diode can be designed that exhibits soft-switching behavior under all conditions relevant in the application, especially also at low current.

5.7.4.5 The Hybrid Diode

Modern fast diodes have been considerably optimized with the described concepts. Thus, there is only a small difference between the CAL diode and the EMCON HE diode in the voltage range of 1200 V with respect to the relationship of forward voltage drop V_F and reverse-recovery charge Q_{RR} [Lut02]. Moreover, there are indications that the limits for possible optimization of fast 1200 V rated diodes based on silicon have almost been approached. However, by parallel- and series connection of different diode designs, these limits given for a single device can be exceeded to some degree.

The hybrid diode [Lut00] is shown in Fig. 5.35a. It consists of a parallel connection of two diodes with contrary switching behavior, whereby on the one hand, a snappy diode D_E is used whose low-doped middle layer is designed as thin as possible; this design is also known as a "Punch-Through" (PT)-diode. On the other hand, a soft-recovery diode D_S is used. With the correct adjustment of the characteristics of both diodes in parallel connection, the low forward-voltage drop of a PT-diode combined with soft-recovery behavior of the soft diode can be obtained simultaneously.

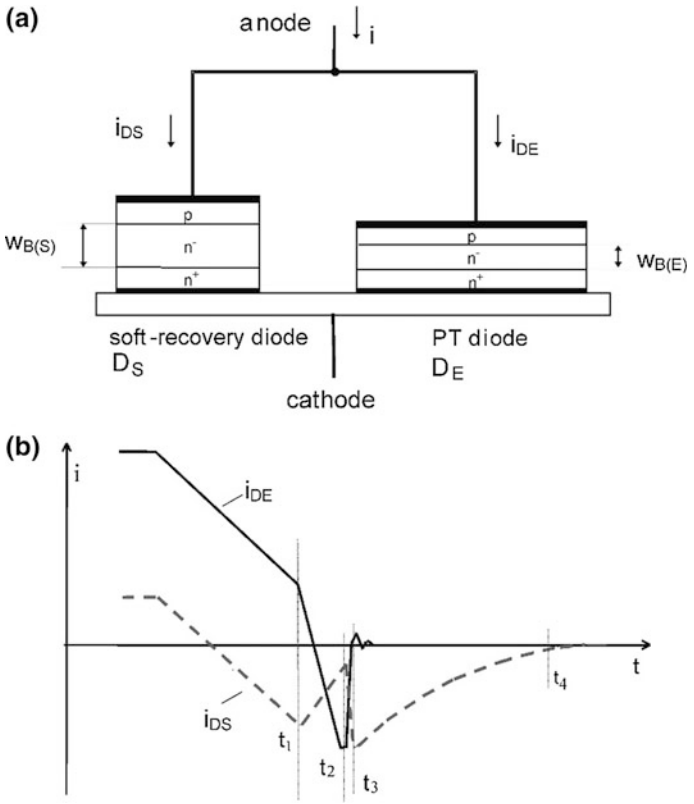


Fig. 5.35 Hybrid diode: **a** Structure; **b** Current waveform in both partial diodes during reverse recovery

The function principle is presented in Fig. 5.35b. The snappy diode D_E carries the main part of the forward current, whereas the soft diode D_S carries a smaller part. At commutation, the current i_{DS} through D_S crosses zero first, and reaches at the instant t_1 its turning point in the reverse current. At this moment, the diode D_E is still carrying forward current. At t_1, the pn-junction of D_S is free of carriers. Then, the diode D_E is commutated with an increased slope di/dt. The total current is still impressed by the outer circuit.

At t_2 the pn-junction of D_E is free, and similarly the reverse recovery current maximum of D_E is reached. Between t_2 and t_3 the hard snap-back of the reverse current i_{DE} in D_E occurs. The reverse current i_{DS} in D_S, which contains still stored plasma, increases with the same slope. The total current as the sum of i_{DS} and i_{DE} shows no reverse current snap-off. Therefore, no high voltage peak is induced. The remaining plasma in the soft-recovery diode D_S is removed between t_3 and t_4. The behavior of the combined arrangement is soft.

For an effective function of the hybrid diode, the diode D_S has to deliver sufficient charge after the reverse-current snap-off in D_E. Thus, the soft diode D_S must be flooded with sufficient plasma, and for this, it must carry between 10 and 25% of the total forward current. The forward voltage drop of both diodes must match to achieve this. In the practical realization, a very fast epitaxial diode with low width w_B of the middle layer is used as diode D_E. Its forward voltage at rated current is in the range of 1.1 V. As diode D_S, a CAL diode with $^1/_6$ of the area of D_E is connected in parallel. This special CAL diode has the double width w_B compared to the Diode D_E, but by implementing a low recombination center density in the base, its forward-voltage drop is adjusted to achieve the desired current distribution between both diodes. D_S features a large tail current, but it is capable of surely overtaking the reverse current snap-off of the epitaxial diode with a six-fold area.

The hybrid diode is proven in application in drives for fork lifts and other electrical vehicles. In these vehicles the battery voltage is typically in the range of 80 V. The applied step-down converters are built up using MOSFETs. To achieve low conduction losses, the MOSFETs must be designed for a voltage not higher than 160–200 V. For the induced voltage peak according to Eq. (5.71),

$$V_{ind} = -L_{par} \cdot \left(\frac{di_r}{dt}\right)_{\max} \tag{5.108}$$

only 80 V margin are available for V_M if the rated voltage of 160 V should not be exceeded for a battery voltage of 80 V. The controlled current is typically high in these applications, i.e. in the range of 200 – 700 A. High currents lead to a large footprint of the modules, and a significant parasitic inductance L_{par} can hardly be avoided. Therefore in Eq. (5.108), the term di_r/dt, the slope of the current after the reverse-current maximum, has to be low to keep the induced voltage in the range that is allowed in such applications. The hybrid diode has proven its capability under these hard requirements.

5.7.4.6 The Tandem Diode

The tandem diode consists of a series connection of two fast diodes in a common housing. An example [IXY00] is shown in Fig. 5.36. The configuration is intended for the application in a step-up converter for Power-Factor-Correction. The concept of the tandem diode aims to offer diodes with a reverse-recovery charge Q_{RR} as low as possible for the application at very high switching frequencies.

Equation (5.37) describes an approximate relationship between the stored charge and the base-width w_B of a diode. The minimal base-width, however, is determined by the required blocking voltage. To express the base-width as function of the desired voltage, we assume the case of a moderate PT-dimensioning, as it was given in (5.15)

$$w_B = \chi \cdot V_{BD}^{\frac{7}{6}} \quad \text{with} \quad \chi = 2.3 \times 10^{-6}\text{cmV}^{-\frac{7}{6}}$$

Inserting this into Eq. (5.37), one obtains

$$Q_F = I_F \frac{\chi^2}{V_{drift}\left(\mu_n + \mu_p\right)} \cdot V_{BD}^{\frac{7}{3}} \tag{5.109}$$

Now the recovery charge of the diodes connected in series, measured as reverse current integral, is equal to the stored charge of a single diode in the series, hence $Q_{RR}^{(series)} = Q_{RR}$. Furthermore, $Q_{RR} = Q_F$ can be assumed as long as the recombination during the reverse recovery is low. However, the total forward voltage of a series of n diodes is

$$V_F = n \cdot \left(V_{drift} + V_j\right) \tag{5.110}$$

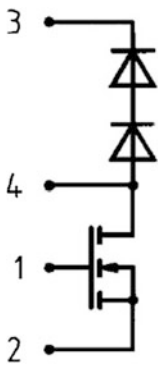


Fig. 5.36 Configuration of a tandem diode and a MOSFET intended for the application as step-up converter

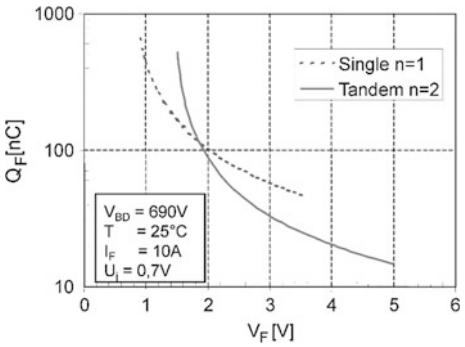


Fig. 5.37 Relation between stored charge and forward voltage for a single diode and for a tandem diode

Expressing the drift voltage of a single diode V_{drift} by the total forward voltage V_F, Eq. (5.109) yields

$$Q_F = I_F \cdot \frac{\chi^2}{(\mu_n + \mu_p)} \cdot \frac{\left(\frac{V_{BD}}{n}\right)^{\frac{7}{3}}}{\left(\frac{V_F}{n} - V_j\right)} \tag{5.111}$$

This relation between stored charge and forward voltage is presented in Fig. 5.37 for a single diode ($n = 1$) and for a tandem diode with $n = 2$. For a forward voltage higher than approx. 1.9 V, the tandem diode shows a better trade-off between V_F and Q_F. The tandem diode recommends itself for applications where very high frequencies and low switching losses are required as the primary feature, but where the current is low and conduction losses are less important. If a forward voltage of 3 V is allowed, a significant reduction of the stored charge can be achieved with the tandem diode. The conduction losses are shared between two devices, which is advantageous regarding the thermal management.

Since a PT-dimensioning was assumed for the partial diodes in the above equations, no soft-recovery behavior can be expected under these conditions. But in the intended applications with voltages of around 600 V, the converters are usually build very compactly, so that low parasitic inductances can be realized. In combination with the typically applied low currents, the soft-recovery requirements can be reduced, also considering that soft recovery leads to switching losses during current decay. In the considered voltage range, no RC-network for symmetrical voltage sharing is necessary using modern devices. Competing with the tandem diode are diodes made from GaAs or Schottky diodes from SiC that are also used for very high frequencies.

5.7.5 *MOS-Controlled Diodes*

The idea underlying the MOS Controlled Diode (MCD) is to improve diode properties by introducing a third electrode, a MOS-gate. Here, we have to anticipate

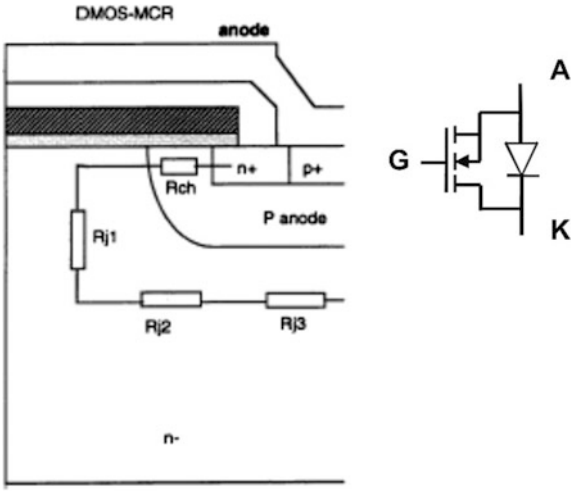


Fig. 5.38 Basic structure of the MOS Controlled Diode (MCD) and equivalent circuit. Figure taken from [Hua94]

some concepts of Chap. 9 about the MOSFET – the Metal Oxide Semiconductor Field Effect Transistor. The basic form of the MCD has the same structure as a power MOSFET. As is shown in Fig. 5.38, the MOSFET contains a pn^-n^+ diode, which consist of the n^+ drain region, the weakly doped n-region and the p-well which is connected with the source metallization, the anode of the diode [Hua94]. The diode lies parallel to the MOS-channel and conducts if the voltage changes its polarity; it is an "anti-parallel" diode. The voltage at the drain electrode is negative in the diode conducting state. The corresponding equivalent circuit is shown in Fig. 5.38 at the right-hand side.

By opening the channel of the MOSFET with a positive gate voltage, a current path parallel to the pn-junction of the diode is created. If the voltage drop along the channel is smaller than the set-in junction voltage of the diode ($\approx$ 0.7 V at room temperature), almost the whole current is flowing through the channel, and the MCD operates in a unipolar mode without carrier injection. Therefore, during commutation no stored charge of injected carriers is to be extracted. Operating in this mode, the structure was called a "synchronous rectifier" [Shm82]. At a channel voltage higher than the set-in junction voltage, charge carriers are injected by the pn-junction. This range of operation, however, is usually excluded when speaking of a synchronous rectifier.

Normally, the MCD is used in a different manner: since the channel is closed during most of the time of forward conduction, the MCD works like a pin-diode. The channel is opened a short time before commutation, so that the pn-junction is nearly shortened via the n-channel. Therefore, the injection by the anode emitter is

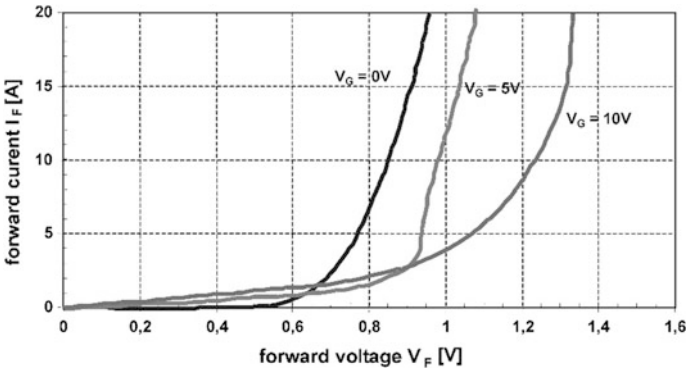


Fig. 5.39 Characteristics of the 1000 V MOSFET IXFX21N100Q in the 3rd quadrant

strongly decreased. This can be expressed also using an effective emitter efficiency of the p-well *and* the channel. Analogously to Eq. (3.96), this can be written

$$\gamma = 1 - \frac{j_n}{j} \tag{5.112}$$

where j_n now denotes the electron current through the channel. If j_n is increased by opening the channel shortly before commutation, γ is strongly decreased and hence also the carrier density is decreased at the anode side of the diode structure. In this way, a drastic reduction of the recovery charge can be achieved. In addition, the inverted carrier distribution leads to a soft-recovery behavior.

The parallel operation of the pin-diode and the MOS-channel is illustrated in Fig. 5.39. A conventional 1000 V MOSFET was found which shows the effect, i.e., the IXFX21N100Q of the manufacturer IXYS. With the channel closed (V_G = 0 V), the current is very small below the set-in junction voltage of approx. 0.6…0.7 V. With opened channel (V_G = 5 V and V_G = 10 V), significant current flows already at smaller voltages where a resistive characteristic is observed. This is caused by the channel resistance, R_{ch}, and resistance components arising from a vertical (R_{j1}) and lateral current paths (R_{j2}, R_{j3}) through the weakly doped n region. On account of R_{ch} the total resistance decreases with increasing gate voltage. This is the operation range of the synchronous rectifier. In the range above 0.7 V, the forward voltage drop at open channel is significantly higher than that in the case of closed channel, as can be seen especially for the higher gate voltage. During the operation cycle of the MCD, the diode works only shortly before turn-off in this part of the characteristics.

In order that opening of the channel results in effective shunting of the current, the voltage drop caused by the external current in the channel and the other resistance components must be smaller than the set–in junction voltage [Hua94]. This holds only up to a critical current I_{crit} given by (at room temperature):

$$I_{crit} = \frac{0.7V}{R_{CH} + R_{j1} + R_{j2} + R_{j3}} \tag{5.113}$$

I_{crit} should be as high as possible and, hence, the resistances R_{Ch} and R_{j1}, R_{j2}, R_{j3} must be as small as possible. An MCD structure optimized for this requirement is shown in Fig. 5.40 [Hua95]. By using a trench structure, the resistances R_{j1} and R_{j2} are removed completely. If additionally the n-region directly under the p-layer is provided with a higher doping, depicted in Fig. 5.40 as n-buffer, also the resistance R_{j3} should be small. But the allowed buffer doping at this position is very closely limited, because it reduces the blocking capability.

The MCD must be switched from the state of fully flooded base, at which no voltage is applied at the gate, to the state of a lower plasma density in the base by applying a voltage to the gate. In this process the main part of the stored charge must be removed. If this is removed by recombination, one has to consider a time of some 10 µs, which is too long for practical application. An n-buffer additionally makes the extraction of holes more difficult. Therefore, an additional p-layer is implemented at the cathode side, which facilitates the extraction of holes to the negatively poled cathode easier.

With these measures, the stored charge before commutation can be reduced significantly; in device simulation even a reduction by a factor of 20–40 was shown [Hua94]. Nevertheless, the stored charge cannot be completely eliminated.

The fundamental disadvantage of all the previously shown variants is that before a blocking voltage is applied to the device, the channel must be closed again. Figure 5.41 depicts an MCD as replacement for a diode in a commutation circuit with an IGBT as switch. The current flows in the freewheeling path, before the

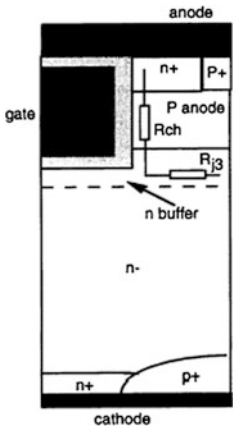


Fig. 5.40 Trench-MCD cell which uses an additional buffer layer at the anode side. Figure from [Hua94]

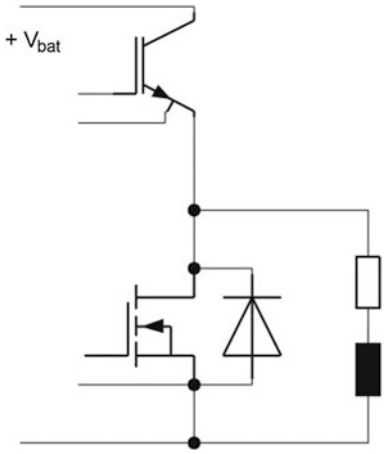


Fig. 5.41 MCD in a commutation branch with IGBT as switching device

diode is commutated from the conducting to the blocking mode at the next turn-on of the IGBT. If the channel of the MCD is open while voltage is applied, the current will not flow into the load but via the channel; a short in the circuit occurs in the commutation branch. Therefore, the channel must be closed beforehand; this point in time must be met with an accuracy of approximately 100 ns. It is hard to ensure this with a driver circuit, however. This disadvantage – that when a positive gate voltage is applied, no blocking capability is given – is the main factor that impedes the implementation of MCD and Trench-MCD in practical applications.

An example for a modified solution without this fundamental disadvantage is the Emitter Controlled Diode (ECD), which is shown in Fig. 5.42 [Dru01]. A low-doped p-zone is connected to the p^+-zone or to the p-well. The highly doped n^+-zone is arranged in the p-well; the path to the low doped p zone is controlled by a MOS-channel. The channel can be prolonged by a very shallow n^+-layer above the low-doped p-layer. Geometry and doping of the respective layers are chosen such that if a positive voltage is applied to the gate and the channel is open, the current takes the path via the low-doped p-layer. For this, the sum of all voltage

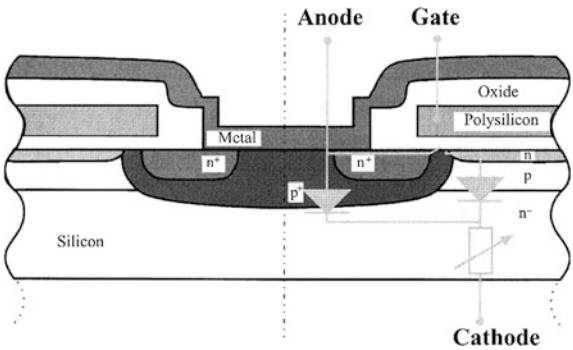


Fig. 5.42 Emitter Controlled Diode (ECD). Figure according to [Dru03]

drops across this path – i.e. the junction voltage at the pn^--junction, the voltage drop in the channel and further resistive parts in the path – must be lower than the junction voltage of the p^+n^--junction.

Without positive voltage at the gate, a $p^+n^-n^+$-structure is given in which a plasma distribution with increasing density to the pn-junction arises, similar to Fig. 5.6. In the case of a channel opened by a positive voltage at the gate, an inverted plasma distribution is given similar to Fig. 5.33 or Fig. 5.31 (right side). The turn-off process is executed only from the open-channel mode; in this mode a turn-off with soft-recovery behavior can be expected. In Fig. 5.43 the internal plasma distribution is compared for both modes. The special progress of the MCD is that, also with open channel, a structure with a blocking capability is given.

The ECD is explained in detail in [Dru01] and in [Dru03] and has not been realized in practice as yet. Nevertheless, it is possible that this idea or similar ideas are used as a foundation for future optimization of diodes in the voltage range of > 3 kV. Even though the necessary effort seems to be high at first glance, one has to consider that, nowadays in applications with IGBTs in this voltage range, the possible switching frequencies are limited especially by the reverse-recovery behavior of the diode. A progress in diodes can lead to an advantage on the system level that might justify the necessary effort.

Derived from the function of the MCD is also the Inverse Injection Dependency of Emitter Efficiency (IDEE) diode [Bab10]. It uses the implementation of n-channels, however there is no gate. The anode layer is highly doped. The therewith achieved high emitter efficiency would cause a snappy reverse recovery behavior, therefore the emitter efficiency is reduced by a parallel electron current. For this, the p-layer is interrupted and a shallow n^+-layer is implemented to ensure an ohmic contact. The structure is shown in Fig. 5.44.

The n-channels are in its dimensioning so narrow that is case of blocking mode the electric field is shielded and is of sufficient distance to the n^+-layers at their

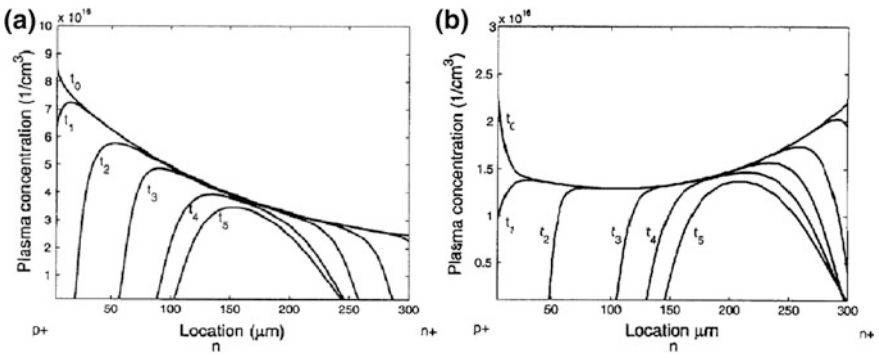


Fig. 5.43 Plasma distribution in the ESD during the reverse-recovery process: **a**) $V_G = 0$, closed channel, **b**) positive gate voltage, open channel. Figure taken from [Dru01] © 2003 IEEE

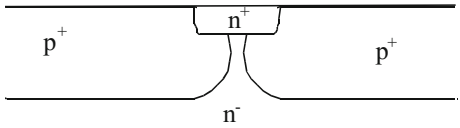


Fig. 5.44 Anode structure of the Inverse Injection Dependency of Emitter Efficiency (IDEE) diode

interface to the contact, see Fig. 5.44. The mode of function is that in forward conduction mode an electron current flows across the channels. This current is limited by the resistance of the channels.

If the voltage drop along the channel and below the p^+-layer, see Eq. (5.113), is larger than the set-in junction voltage of the pn-junction of the diode (≈0.7 V at room temperature), the pn-junction will inject carriers. In this rough simplification, the potential at the junction remains at the junction voltage and electron current across the channel is given by

$$j_n(x_p) = \frac{V_{bi}}{R_{CH} \cdot A_{CH}} \tag{5.114}$$

with R_{CH} as resistance of the channel and A_{CH} as its area. It remains the same even if the current density of the diode is increased. Therefore, at increased current density the share of electron at the total current is reduced, and with this the emitter efficiency γ according to Eq. (5.112) should increase. For usual pn-junctions this is opposite: The emitter efficiency is reduced with increasing current density, see Sect. 3.4. If Eq. (3.96) is rewritten for a p-emitter, it reads

$$\gamma = 1 - j_n(x_p)/j = 1 - q \cdot h_p \cdot p_L^2/j \tag{5.115}$$

Since p_L is approximately proportional to j and is contained with the power of 2 in Eq. (5.115), it results in a decrease of γ proportional to j.

Additionally, Fig. 5.8 has shown a decreased effective lifetime τ_{eff} with increased carrier density, caused by emitter recombination which is stronger with higher carrier density.

The dependency of the emitter efficiency on the current extracted from the simulation results is shown in Fig. 5.45. The conventional p-doping controlled diode (reference) shows the typical strong decay of γ with increasing currents. The IDEE-diode shows low γ at low current. The desired inverted dependency of γ with the current density is not completely, however in tendency achieved. The IDEE diode shows increasing values of γ at high currents.

The IDEE-diode gives a potential of further improvement of the reverse-recovery behavior at low current densities as well as the possibility of lower forward voltage at high current. This leads in the same way to a higher surge-current capability. Surge current behavior is discussed later in Sect. 12.2.

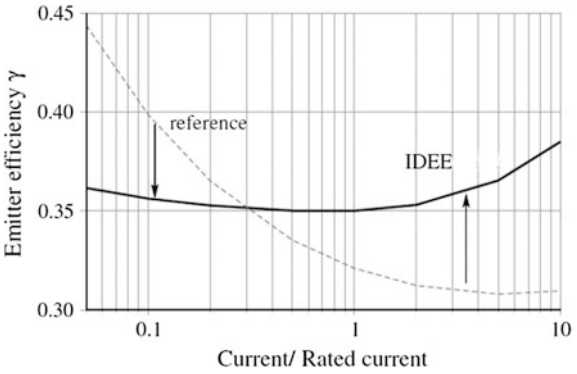


Fig. 5.45 Emitter efficiency γ of the anode-side emitter depending on forward current for a usual diode with homogeneous p-layer (reference) and for the IDEE diode. Figure from [Bab10] © 2010 IEEE reprint with permission

5.7.5.1 Diodes with Cathode Side Hole Injection

While all measures up to now modified the anode side, it was found that also the cathode side, the nn$^+$-junction, can be used to improve the reverse recovery behavior. Structures which inject at reverse recovery additional holes from the cathode side are the Field Charge Extraction (FCE) structure [Kop05] and the Controlled Injection of Backside Holes (CIBH) structure [Chm06]. Since there are additional p-layers at the cathode side, no electric field can build up between the plasma and said layers. It can be achieved that the plasma does not detach from the cathode. Instead of Eq. (5.93), $|v_r| = 0$ applies and from Eq. (5.94) results $w_x = w_B$. This extends the voltage range of soft recovery. Moreover, it was shown for the CIBH-diode that, if at the end of the reverse recovery process the space charge punches to the n$^+$-layer (see Eq. (5.104)), the backside p-layers inject additional holes and damps possible oscillations [Fel08].

Since these structures were developed with the intention to increase the capability of the diode to withstand dynamic avalanche, they will be described in more detail in Sect. 13.4.

5.8 Outlook

For the soft-recovery behavior of fast diodes, appropriate solutions have been found in the voltage range smaller than 2000 V. Even though further optimization is possible, there are indications that the design is already approaching the limits of what is possible for pin-diodes in silicon. Still there is a potential of hybrid structures for improving diodes.

For the voltage range of 3000 V and more, considerable work still has to be done to realize diodes with satisfactory reverse-recovery behavior in applications with very high power. These applications combine switching slopes, which are much steeper than formerly occurring using thyristors and GTOs, with a significant parasitic inductance in the commutation circuit. For these applications the devices

have to be optimized to avoid voltage peaks and oscillations even under such hard conditions. Furthermore, dynamic ruggedness is very important; this will be explained in Chap. 12.

Regarding applications with high switching frequencies, Schottky diodes are the better choice. Schottky-diodes from GaAs are available as single diodes for 300 V, and they may be developed to reach the voltage range of 600 V. Though Schottky diodes in SiC have already been established for the range of 600 V and 1200 V (see Chap. 6), they can also be designed for a higher voltage range. Because of problems involving material quality and defects, SiC devices are still available only with comparatively small area. For applications in the wide field of motor drives, they should become available in sufficient area (of up to 1 cm^2) to carry enough current per device and at costs that are competitive to silicon devices.

Moreover, there are encouraging results in research regarding pin-diodes in SiC; in particular, here blocking voltages far above 10 kV are possible with a single diode. First results seem to show that the physics of reverse recovery, which was previously explained, can be used in a similar way for analysis and optimization of SiC pin-diodes [Bar07]. However, applications in the high-power range require high currents and, thus, large device areas too.

Therefore, diodes from silicon will probably dominate the market for a long time, and still further work for optimizing them is necessary. In the field of fast diodes, Si and SiC will most likely exist in parallel for some time.

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